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Unit D1 - Maritime transport and logistics

Guidance on verification for the FuelEU Maritime Regulation

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Disclaimer

This document is part of a series of documents provided by the Commission services for supporting the implementation of FuelEU Maritime Regulation. The guidance represents the views of the Commission services at the time of publication. It is not legally binding.

The aim of this guidance is to support the implementation of the [FuelEU Maritime Regulation](#), by explaining its requirements in a non-legislative language. For some more specific technical issues, further guidance documents may be made available in the future. Such guidance documents would be accessible through the [Commission's website](#) as well.

This guidance document was endorsed in its first version by Member States of the ESSF Subgroup on the Implementation of the Fuel EU Maritime Regulation in its meeting on 25th June 2026.

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1 Scope of the document - verification items specific to FuelEU

This guidance focuses exclusively on the monitoring, reporting, and compliance elements subject to verification that are specific to the FuelEU Maritime Regulation ([Regulation \(EU\) 2023/1805](#)). The main reference to interpret the FuelEU maritime regulation is the legal text and its associated secondary legislation. This Guidance complements the regulation with practical recommendations on how to implement the regulation. For the technical implementation of FuelEU Regulation, where the legislative framework or the applicable guidance documents do not provide specific provisions, relevant IMO resolutions and other applicable EU and internationally recognized standards should be used as supplementary technical references.

Due to the high degree of coherence in verification principles between FuelEU and the EU MRV/ETS frameworks, this guidance does not duplicate verification elements that overlap with those frameworks, as these are already comprehensively addressed in the respective Commission guidance documents for those frameworks (see Text box 1). Likewise, aspects related to verifier accreditation fall outside the scope of this document.

Text box 1 – Overview of existing guidance relevant to certain aspects of FuelEU verification

The [EU ETS and MRV Maritime Guidance for Accreditation and Verification](#) (GD3) already provides detailed verification guidance for the EU MRV/ETS, including underlying verification principles which remain valid for FuelEU. Although there are distinct legal acts on verification activities for EU MRV/ETS ([Commission Delegated Regulation EU 2023/2917](#)) and FuelEU ([Commission Implementing Regulation EU 2024/2027](#)), the verification objectives and principles are highly comparable. The process and standards of accreditation are also very similar, although formally distinct.

In addition, the [EU ETS and MRV Maritime General guidance for shipping companies](#) (GD1) explains the monitoring and reporting concepts and definitions of those frameworks - many of which also apply to FuelEU.

Moreover, some FuelEU verification elements not covered by GD1 or GD3 have already been addressed in the existing [FuelEU guidance document for shipping companies](#).

This document is structured around the three main stages of the FuelEU verification process:

1. Assessment of FuelEU monitoring plans;
2. Verification of FuelEU reports
3. Verifier role in ensuring correct application of flexibility mechanisms

Each chapter begins with a table showing an inventory of verification elements identified as unique to FuelEU. Where relevant guidance already exists in other documents, cross-

references are provided. For verification elements not yet covered elsewhere, new guidance has been developed and presented here as appropriate.

The guidance provided herein represents recommended approaches and interpretations to assist verifiers in applying legal requirements consistently. It is non-binding and intended as a reference document, offering examples and suggestions on how verifiers might perform their activities. Verifiers, as accredited entities with sector-specific expertise in data and information auditing, may decide to take additional, or different, measures to reduce verification risks to an acceptable level to obtain reasonable assurance that the FuelEU monitoring plan, FuelEU report and/or the use of flexibility mechanisms are in conformity to the FuelEU Maritime Regulation.

Note that this document as well as the relevant legislation and cited reference documents can all be subject to updates. The reader is thus recommended to always consult the most recent versions of the documentation which are primarily available from the websites of DG MOVE¹ and DG CLIMA².

Text box 2 briefly reiterates key verification principles that are common to both FuelEU and the EU MRV frameworks. Recommended further reading can be found in chapter 3 “verification principles” of [EU ETS and MRV Maritime Guidance for Accreditation and Verification](#). (Consult DG CLIMA website for latest version).

Text box 2 – Key principles of verification for the EU MRV, ETS and FuelEU

Independence of the verifier:

FuelEU Article 12 states that the verifier shall be independent from the company or from the ship operator and shall carry out the activities required under this Regulation in the public interest. The overall goal of verification is to ensure reported data is reliable, credible, accurate and complete. Verifiers are expected to apply knowledge and experience of data and information auditing³ to ensure reported data are coherent and free from inconsistencies.

Professional scepticism:

A verifier has to carry out his/her activities in a sound and objective professional manner. This implies for example that the verifier should not automatically accept evidence obtained during verification, but should analyse this evidence thoroughly in line with the required level of assurance.

Reasonable assurance:

According to FuelEU Article 16, a verification report should only be issued where the verifier has reasonable assurance that a FuelEU report is free from material misstatements or non-conformities. “Reasonable assurance” means for the MRV and FuelEU a high – but not absolute – degree of assurance as to whether

¹ See https://transport.ec.europa.eu/transport-modes/maritime/decarbonising-maritime-transport-fueleu-maritime_en

² See https://climate.ec.europa.eu/eu-action/transport-decarbonisation/reducing-emissions-shipping-sector_en

³ See EU 2024/2027, Article 32(b) “competence requirements for shipping FuelEU auditors and shipping FuelEU lead auditors”

the FuelEU/MRV emissions report are free of material misstatements⁴. In both the MRV and FuelEU verification acts, verifiers are required to take measures to reduce the “verification risk” to an acceptable level to obtain reasonable assurance that monitoring plans/reports are in conformity to the respective regulations. Verification risk means, in both cases, the risk, as a function of inherent, control and detection risk, of the verifier expressing an inappropriate verification opinion where the FuelEU/MRV report are not free of material misstatements;

Materiality:

Only FuelEU reports that are free of material misstatements can be regarded as satisfactory.

According to Article 2(4) and 18 of Implementing Regulation (EU) 2024/2027, the cut-off point (materiality level), above which the verifier should consider misstatements, individually or taken together with other misstatements to be material, is 5% for – in particular – the following reported totals:

- Fuel and electricity consumption
- Distance travelled
- Time spent at berth and at sea

2 Verifier assessment of FuelEU monitoring plans

Since FuelEU has different policy objectives than the MRV framework, shipping companies are required to monitor several different parameters through procedures implemented either on board or ashore. These extra monitoring parameters are reflected in the [FuelEU monitoring plan template](#) which contains several additional tables as compared to the [MRV monitoring plan template](#)⁵.

The additional FuelEU procedures/parameters are listed in column 1 of Table 1 below while columns 2 and 3 show the availability of existing guidance in other documents. Column 4 lists the corresponding section of this document where additional guidance can be found.

The following sections provide additional guidance on how verifiers may assess these FuelEU-specific procedures/parameters to ensure a ship has a satisfactory FuelEU monitoring plan. In accordance with Article 5(3) of Implementing Regulation (EU) 2024/2027, verifiers may employ methods such as inquiry, document inspection, observation, or any other audit techniques considered appropriate for assessing the monitoring plan.

The information contained in a FuelEU monitoring plan is relevant to the verifier both when assessing the monitoring plan as well as the eventual verification of a ship’s FuelEU report. Consequently, there is a degree of interplay between the guidance presented in Chapters 2 and 3 of this document but guidance is not repeated between the two.

⁴ According to Article 2(6) of <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32023R2917> and Article 2(3) of <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R2027>

⁵ Note that there are also various MRV parameters/procedures that are not required for FuelEU, such as recording the amount of cargo carried and number of passengers

Guidance on verification for FuelEU Maritime Regulation

Table 1 – Additional tables in FuelEU monitoring plan template, as compared to MRV. Note: organised by theme and not necessarily in the order of the template

FuelEU specific monitoring parameter / procedure and reference in monitoring plan template (Annex I (EU) 2024/2027)	(Partially) addressed in CLIMA GD1 or GD3?	(Partially) addressed in FuelEU guidance for shipping companies?	Location of additional guidance in this document (section)
OPS- Table B.4 equipment to allow connection to OPS	No	Yes, chapter 6 + Annex II	2.1.1
OPS – Table B.6: Established total electrical power demand of the ship at berth	No	Yes, chapter 6 + Annex II	2.1.2
OPS: - C.1.6 Procedures for monitoring energy provided by OPS	No	Yes, chapter 6.1.4	2.1.3
OPS – D.2: <u>In case of data gaps</u> : Methods to be used to estimate energy consumption from OPS and ZET	No	Partially, Annex II	2.1.4
Time spent at berth. – Table C.3.4 and Table D.5 (concerning methods to treat data gaps regarding time spent at berth)	Time at berth not monitored/reported under MRV.	Only partially in Annex II	2.2
Zero Emission Technologies – Table B.5: Description of Zero Emission Technology	No	Yes, chapter 6.2	N/A Pending acceptance criteria for ZETs
Wind Assisted Propulsion Systems – Table B7: Description / Table B8: List of consumers	No	Yes, chapter 4.5.	2.3
Substitute sources of energy – Table C.2.1 (description of energy consumption measurements)	No	No	2.4
WtW emission factors – Table B9: Procedures for monitoring and reporting the WtT and TtW emission factors of energy to be used on board in accordance with the methods specified in Article 10 and Annexes I and II	Partially, GD1 Annex III on “zero-rating” TtW for ETS. PoS / PoC documentation also explained.	Yes, chapter 4	See existing guidance for shipping companies
Ice class - Table B1: identification of ice class	Yes	Yes – Annex II.]	See existing guidance for shipping companies
Ice class: monitoring fuel consumption, time and distance travelled in ice conditions- C.1.8: Procedures for monitoring fuel consumption of each fuel consumer when sailing in ice conditions. - Table C.3.3: date, time and position when entering and leaving the ice conditions, the distance travelled when sailing in ice conditions	No, ETS only has exemption for ice-class with technical characteristics not sailing in ice. Time and distance through ice is only a voluntary parameter for MRV.	No specific guidance is given for monitoring fuel consumption, time or distance in when sailing in ice conditions	2.5

2.1 Verification of onshore power supply

The following sub-sections first briefly discuss the requirements of the tables in the FuelEU monitoring plan related to OPS. Guidance is then given as to how verifiers might assess the completeness, accuracy, relevance and conformity of procedures written in a FuelEU monitoring plan.

Note: For Ships that are required to use OPS to cover all electrical power demand at berth, they are not obliged to switch off fuel-consuming equipment at ports for which no electrically powered redundant equivalent exists onboard — for example, thermal boilers fired by gas or oil. The thermal boilers should however be detailed in Table B.3 of the FuelEU monitoring plan⁶ and their fuel consumption, including at berth, reported for GHG intensity calculations, covering energy used in EU ports. Electric boilers, where installed, are also recommended to be included in Table B.3 with a description of the electrical capacity.

2.1.1 Equipment to allow connection to OPS

Table 2 – FuelEU monitoring plan template – **Table B.4**

Equipment to allow connection to on-shore power supply			
OPS equipment reference no.	OPS equipment (name, type)	Technical description of OPS equipment specified voltage and frequency, including the gear specified in IEC/IEEE 80005-1 (High Voltage) and IEC/IEEE 80005-3 (Low Voltage) and position onboard	Standards (IEC/IEEE 80005-1 (High Voltage) and IEC/IEEE 80005-3 (Low Voltage)).

According to the FuelEU monitoring plan template, Table B.4 is mandatory for container ships and passenger ships only, since these are the ship types subject to the zero-emission requirements for energy used at berth as per Article 6 of FuelEU.

However, verifiers should also consider this table mandatory for any ship which intends to use OPS throughout the reporting year (even those using OPS on a voluntary basis).

Assessing a monitoring plan for OPS elements prior to 2030

As it is the case that the requirements of FuelEU Article 6 apply only from 1 January 2030, verifiers might assess monitoring plans for any ship type as satisfactory without completion of Table B.4 – provided that the monitoring plan is updated prior to 2030, for those ships obliged to connect to OPS from 2030.

To ensure that the monitoring plan is updated ahead of reporting year 2030 for passenger and container ships, the verifier could – for instance – note the required update in the internal verification documentation⁷, or approve the monitoring plan as satisfactory but with an open comment for this table to be updated ahead of 2030. Following the verification of a FuelEU report prior to 2030, the verifier

⁶ See template in Implementing Regulation (EU) 2024/2031

⁷ According to article 2(12) of Implementing Regulation 2024/2027, “internal verification documentation” means all internal documentation that a verifier has compiled to record documentary evidence and justification of activities carried out to assess the monitoring plan or verify a FuelEU report and partial FuelEU report

could also issue a recommendation for improvement to the shipping company noting that OPS equipment needs to be added to the monitoring plan prior to 2030⁸. Recommendations for improvement are notably part of the Verification Report⁹ which is recorded on the FuelEU database. In parallel, shipping companies are required to review their monitoring plans annually, and the FuelEU Database¹⁰ system prompts them to complete this process. Together, these mechanisms can ensure that monitoring plans are reviewed and updated in a timely manner ahead of the enforcement of OPS obligations for container and passenger ships in 2030.

Assessing the appropriate technical specification for the ship:

Table B.4 of the FuelEU monitoring plan template requires to specify if the ship's onboard OPS connection point complies with the following technical specifications:

1. IEC/IEEE 80005-1 (High Voltage)
2. IEC/IEEE 80005-3 (Low Voltage)

The above are, according to the IEC/IEEE, specifications for the design, installation and testing of high/low-voltage shore connection systems, onboard ships and on shore¹¹. However **FuelEU itself does not directly require a ship's OPS connection point to adhere to the above technical specifications**. For this reason, a verifier might – at their discretion – assess a monitoring plan as satisfactory when a ship's OPS connection point adheres to different standards.

It is nonetheless important that **as soon as article 6 of FuelEU is in application from 2030, there are implications for container and passenger ships with onboard OPS equipment which does not adhere to established technical specifications for seagoing ships**. In particular, FuelEU article 6(5)f permits exemptions to OPS on the grounds of non-compatibility with shore installations in port only “provided that the installation for shore-connection on board the **ship is certified in accordance with the technical specifications set out in Annex II to Regulation (EU) 2023/1804** for the shore-connection systems of seagoing ships.”

Annex II to Regulation (EU) 2023/1804 (The Alternative Fuels Infrastructure Regulation, or AFIR) presently¹² provides only 1 technical specification for shore-side electricity supply for seagoing ships being:

- IEC/IEEE 80005-1:2019 / AMD1:2022 for high-voltage shore connections

Therefore, while a verifier may still assess a FuelEU monitoring plan as satisfactory for passenger and container ships equipped with an OPS connection point that does not conform to the above technical

⁸ According to article 22 of Implementing Regulation 2024/2027, verifiers can issue Recommendations for Improvement to shipping companies in relation to uncorrected misstatements and non-conformities not leading to material misstatements after the verification of FuelEU reports. In subsequent verification periods, the verifier should then assess whether and how the company has implemented those recommendations for improvement.

⁹ According to the Verification Report template in Annex II of Implementing Regulation 2024/2027

¹⁰ The primary foundational structure upon which the FuelEU database is built is THETIS-MRV. For further information consult

Commission Implementing Regulation (EU) 2026/394.

¹¹ A description of the scope of these standards can be found in Table 5.3 of [EMSA Guidance on Shore-Side Electricity Part 2](#)

¹² Note that a review of the Alternative Fuels Infrastructure Regulation is, according to Article 24 of the AFIR Regulation, required by 31 December 2026

specification (or to an alternative specification established in a future revision of the AFIR), the potential implications for the ship from 2030 onwards should be documented by the verifier — for instance, within the internal verification documentation.

When assessing FuelEU monitoring plans, verifiers might also consider whether a stated OPS technical specification is plausible for the ship:

- While Table B.4 of the template of the monitoring plan does provide IEC/IEEE 80005-3 (Low Voltage) as an option, verifiers should use their professional judgment to assess whether this standard is appropriate for the ship type to enable shore electricity consumption in practice. Pending a review of the AFIR to potentially include a reference to 80005-3, an OPS connection point adhering to this standard would today render the ship unable to make use of the above-mentioned OPS exemption¹³. The electrical power demand of large passenger and containerships might also exceed the electrical demand that this low voltage standard is intended for.
- According to the IEC, 80005-3¹⁴ (Low Voltage) is applicable to ships requiring up to 1 MVA while at berth (which would be in the region of 1MW, depending on the power factor applied). Although there may be some ships in scope of FuelEU for which this standard is appropriate, others might have a too large power demand. Indeed the average power demand of container and passenger ships > 5 000 GT (those in scope of FuelEU article 6) is considered to be at least 1.5 MW and far higher for larger ships including cruise passenger ships¹⁵.

For ships in scope of FuelEU which are not passenger or containerships, there are no obligations to connect to OPS at berth on the basis of FuelEU article 6. However, the voluntary connection to OPS is an option for other ship types. In cases where ships are equipped with onboard OPS connection points, it is required that the ship's monitoring plan is assessed by the verifier with Table B.4 completed prior to connections at OPS on port calls in the scope of FuelEU.

FuelEU does not in any way prescribe particular technical specifications for the onboard OPS connection point for ships other than container or passenger ships. Nonetheless, when assessing the FuelEU monitoring plan for such ships, verifiers are recommended to ensure the ship's OPS connection point adheres to recognised international standards (such as those mentioned above or in Table 5.3 of [EMSA Guidance on Shore-Side Electricity Part 2](#)) or, at minimum, a confirmation that the installation has been accepted in accordance with the relevant rules of the ship's classification society or flag state.

In general, verifiers might confirm the adherence of a ship's onboard OPS equipment to the above technical specifications via consultation of the following documentation (in no particular order) which they could request from the shipping company¹⁶:

- OPS equipment supplier declarations of conformity to the corresponding IEC/IEEE 80005 standard
- Class / statutory accepted approval certificates for OPS equipment
- OPS system drawings, descriptions, or circuit diagrams.

¹³ Unless Annex II to the AFIR is revised prior to 2030.

¹⁴ <https://webstore.iec.ch/en/publication/30632>

¹⁵ EMSA SSE/OPS Quick-Reference Guide <https://www.emsa.europa.eu/publications/inventories/download/7257/4799/23.html>

¹⁶ Article 4(3) of EU 2024/2027 states that companies shall provide the verifier with information before the start of the assessment of the plan. This includes documents of the ship's installations as well as any other information the verifier deems relevant.

- For ships classed by an IACS member, IACS Recommendation No. 182 (Dec 2024) states the latter documentation should be available onboard the ship.
- The IMO interim guidelines on safe operation of OPS (MSC.1/Circ. 1675) also recommend the above documentation is kept onboard, together with OPS procedures in the ship's Safety Management System.
- Classification certificates
 - It is noted that certain classification societies have additional class notations for OPS for which adherence to technical specifications like IEC/IEEE 80005-1 is a prerequisite. In such cases, the verifier could request confirmation of the rules of eligibility for a certain notation as well as evidence (via a valid class certificate) that the ship holds the notation.

Position of OPS equipment onboard:

According to Table B.4 of the FuelEU monitoring plan template, the description of the ship's equipment to allow connection to OPS supply should also state the position of the equipment on board.

It is recommended the position be indicated in at least the following broad terms on the monitoring plan:

- Side: Port / Starboard
- Length: Forward / Midship / Aft
- Height: at which level above surface of the sea in a certain weight condition.

Stating the position of the OPS connection point on the monitoring plan could support a verifier to corroborate the existence of this equipment via ship drawings or documentation when assessing the monitoring plan.

The position of a ship's OPS equipment may also be relevant to enable a verifier to corroborate the actual use of OPS by the ship in certain ports or terminals. For example, in certain berths within a terminal, OPS may only be possible in certain mooring/berthing/cargo-transfer configurations which could be corroborated by for instance, ship logbook entries, AIS data, and OPS information from the port authority. In this sense, information about the position of OPS equipment onboard could be relevant for enforcement purposes too.

Verifiers might corroborate the position of a ship's OPS equipment via the following, non-exhaustive, sources of information which they could request:

- Site visits, including onboard the ship if deemed appropriate by the verifier
- OPS procedures in the ship's Safety Management System
- Class / flag approved ship drawings or documentation which show the connection point, for instance: electrical drawings, general arrangement plan, etc.

2.1.2 Established total electrical power demand of the ship at berth

According to the FuelEU monitoring plan template, Table B.6 is mandatory for container ships and passenger ships as follows:

Table 3 – FuelEU monitoring plan template – **Table B.6**

Established total electrical power demand of the ship at berth	
Established total electrical power demand of the ship at berth (kW)	Data Source

“Established total electrical power demand of the ship at berth” means – according to FuelEU article 3(26) – the highest value, expressed in kilowatts, of the total demand for electricity of a ship at berth, including hotel and cargo handling workloads. This should also include cargo-handling and other relevant operational electrical consumers.

The value is primarily important for the functioning of the Regulation in order to, from 2030, enable the calculation of penalties for non-compliance to article 6 for container and passenger ships (as per FuelEU article 23(5)). The value is intended to act as a disincentive for non-compliance to the OPS requirements of article 6 for container and passenger ships. Accordingly, the legislators emphasised that it is the **highest value**, expressed in kilowatts, of the total demand for electricity of a ship at berth, including hotel and cargo handling workloads. This is important because a higher value results in a higher penalty for non-compliance to the OPS requirements¹⁷.

The “established total electrical power demand of the ship at berth” is also important in the following secondary ways:

- For the exchange of information prior to entry into ports. FuelEU article 6(8) requires ships that intend to connect to OPS to indicate in advance the amount of power they expect to require during that port call¹⁸.
- As a reference point for verifiers during the verification of a FuelEU report. For instance, to ascertain that the amount of energy claimed by ships to have been received from OPS is plausible in the event of data gaps or incomplete documentation¹⁹.

FuelEU article 8(e) provides companies two methods which may be used to determine the ship’s established total electrical power demand (in kW), being:

- **Method 1 (Default):** the value as stated on a ship’s electrical load balance or electrical load study used to demonstrate compliance with Regulations 40 and 41 of SOLAS Chapter II-1, as approved by the ship’s flag state or RO.

¹⁷ Since the penalty is a function of 1.5 EUR/kWh x the established total electrical power demand of the ship at berth x hours spent at berth in non-compliance.

¹⁸ This article does not state that ships must report their established total electrical power demand, only their expected demand. However stating the maximum may nonetheless be useful.

¹⁹ In such a plausibility check, verifier could for example check the value provided by the ship for monitoring plan table B.6 and the number of hours the ship reported to have been at berth.

- **Method 2 (only applicable if method 1 is not possible):** 25% of the total of the maximum continuous ratings (MCR) of the main engines of the ship as specified in their Engine International Air Pollution Prevention (EIAPP) certificate or, if the engines are not required to have an EIAPP certificate, on the nameplate of the engines.

Method 1 is intended to be the primary option. Method 2 is only intended to be used when companies are not able to provide a reference for the ship in accordance with Method 1. The following discusses these 2 methods in more detail.

2.1.2.1 Method 1: kW value as stated on a ship's electrical load balance or electrical load study

Regulation 40(1) of SOLAS Chapter II-1 requires, for all ships, that electrical installations are such that:

1. all electrical auxiliary services necessary for maintaining the ship in normal operational and habitable conditions will be ensured without recourse to the emergency source of electrical power
2. electrical services essential for safety will be ensured under various emergency conditions; and
3. the safety of passengers, crew and ship from electrical hazards will be ensured.

With respect to the above, Flag Administrations are expected to take appropriate steps to ensure uniformity in their implementation and application – in particular by referring to the recommendations published by the International Electrotechnical Commission (IEC).

Meanwhile, Regulation 41 of SOLAS Chapter II-1 concerns the ship's main source of electrical power and lighting systems. It includes the requirement that the main source of electrical power consists of at least two generating sets and should be of sufficient capacity to supply all those services mentioned in regulation 40.1.1

As such, it follows from Regulation 40 and 41 of SOLAS Chapter II-1 that an electrical load balance or study would – in some form – be required to demonstrate compliance of the above to Flag Administrations or ROs. However, neither of these SOLAS Regulations specify exactly that a formal load balance / study must be carried out, nor is there any specified format or template that this study should follow. This may have the following implications for the verifier when determining which value to accept as the ship's 'established total electrical power demand of the ship at berth':

- the ship may not have an (flag/RO approved) electrical load balance / study if the flag / RO accepted compliance without requiring this documentation
- the ship has a load balance/study from a certain period (e.g. newbuilding while the ship was registered with a certain flag). In the meantime, the ship has changed flag and installed new electrical systems or engines and the load balance / study was not required by the new flag / RO to be updated.
- The ship has a load balance / study but there is no value that clearly corresponds to the definition of 'established total electrical power demand of the ship at berth' used for FuelEU.
- The electrical load balance / study may for instance provide a value which includes electrical demand for the ship including during propulsion at sea (such as electrically driven thrusters, pumps, steering gears etc) whereas what is required for FuelEU is the electrical demand **at berth**, including hotel and cargo handling workloads;

A recommended approach is therefore as follows:

1. Where ships do have an electrical load balance, study or similar document which functions as evidence of compliance to Regulation 40 and 41 of SOLAS Chapter II-1, this documentation should be provided to the verifier together with evidence of its approval from the RO or flag state (such as a stamp on the document or a letter of approval).
2. The verifier should check whether the provided documentation states a value which corresponds to the FuelEU definition of “established total electrical power demand of the ship at berth” (which again means: *the highest expressed value of the total demand for electricity of a ship at berth, including hotel and cargo handling workloads*)
 - a. If there is a value corresponding to the above definition for the electrical demand at berth, this should be used
 - b. If there are multiple plausible but not exactly corresponding values for the electrical demand at berth, the highest value in kW should be used
 - c. If there is no specific value for the ship’s total electrical power demand at berth including hotel and (where applicable) cargo handling workloads, the sum of the following values should be used, where they are available in the provided documentation, in accordance with the classification given in IACS Unified Interpretation UI SC134:
 - i. Secondary Essential Services²⁰
 - ii. Services for habitability²¹

In cases where the ship has no electrical load balance / study or similar document, or where none of the values given in the document correspond to the definition given in FuelEU article 3(26) or the guidance above, method 2 should be used.

2.1.2.2 Method 2: kW value considered to be 25% of the total of the MCR of the main engines of the ship

MARPOL Annex VI Regulation 13 on the control of Nitrogen oxides (NO_x) requires – via the NO_x Code – that engines²² installed on a ship are certified via an Engine International Air Pollution Prevention (EIAPP) certificate. The template for this certificate as provided in Appendix 1 of the NO_x Code includes a field for the “rated power” which means the maximum continuous rated power output as specified on the nameplate and in the Technical File of the marine diesel engine²³. This EIAPP engine certificate is required to remain onboard at all times.

Meanwhile, MARPOL Annex VI Regulation 6 requires ships to carry an International Air Pollution Prevention Certificate (IAPP) as issued by their flag administration / RO. The IAPP certificate follows a standard format set out in Appendix 1 of MARPOL Annex VI and includes a supplement which should be permanently attached to the IAPP certificate. The supplement lists all of the engines installed on the ship

²⁰ According to IACS UI SC134, Secondary Essential Services are those services which need not necessarily be in continuous operation to maintain propulsion and steering but which are necessary for maintaining the vessel's safety.

²¹ According to IACS UI SC134, Services for habitability are those services which need to be in operation for maintaining the vessel's minimum comfort conditions for the crew and passengers (e.g. heating, mechanical ventilation, etc)

²² with a power output of more than 130 kW

²³ According to 1.3.11 of the NO_x Technical Code

including parameters such as the manufacturer and model, serial number and “use”. The term “use” in most cases signifies whether the engine is used for main propulsion, or auxiliary or emergency power²⁴. Method 2 therefore involves summing the kW rated power of a ship’s main engines and assigning the ship’s established total electrical power demand as 25% of this value. Main engines included in this calculation must always comprise every engine listed as “main engine” on the ship’s FuelEU and MRV Monitoring Plan (in both cases, table B.3)

The following are recommendations to follow:

- Any engine listed as “main engine” on the ship’s EIAPP or IAPP certificate but not expressed as such in table B.3 of the FuelEU/MRV monitoring plan should be added to table B.3 and included, for FuelEU, in the calculation of the ship’s established total electrical power demand of the ship at berth.
- In case of discrepancies between the rated power of the engine on the EIAPP, the IAPP or technical description for the consumer unit in table B.3, the highest value should be used in the calculation of the ship’s electrical demand at berth.

The rated power of the engine should only be sourced from the nameplate on the engine in cases where the engine is not required to have an EIAPP. This should – in most cases – concern only engines with a rated power of less than 130 kW.

Note on Engine Power Limitation:

It should be noted that the 25% of MCR figure is solely a proxy used to estimate the ship’s electrical power demand at berth in the absence of an electrical load balance/study, and bears no relation to any power limitation of the main engines. As such, any engine power restriction has no bearing on this calculation, and whenever the source of this information contains two values, the highest should be used.

2.1.3 Procedures for monitoring energy provided by OPS

The FuelEU monitoring plan template requires a procedure to be developed for monitoring energy provided by OPS (table C.1.6 in the monitoring plan template, see below). Related to this, the company is also required to state a method to estimate energy consumption from OPS in the event of data gaps (table D.2 in the monitoring plan template) which is discussed separately in section 2.1.4.

Table 4 FuelEU monitoring plan template – **Table C.1.6**

Procedures for monitoring energy provided by on-shore power supply (OPS) (where applicable)

Title of procedure	Monitoring energy provided by on-shore power supply (OPS)
Reference to existing procedure	
Version of existing procedure	

²⁴ “Use” refers to the applicable test cycle for the engine which, according to 3.2 of the NOx Code, are different depending on the type of engine e.g. main propulsion engines, auxiliary engines, etc

Description of procedure (a brief description of the procedure can be provided if already existing outside the monitoring plan)	
Name of person or position responsible for this procedure	
Location where records are kept	
Name of IT system used (where applicable)	

Monitoring OPS consumption via Electricity Delivery Notes (default method):

The primary procedure ships are expected to follow in order to monitor energy from OPS is via Electricity Delivery Notes (EDNs). This is because EDN's are specifically defined in Annex I of the FuelEU Regulation in the context of the methodology for establishing GHG intensity of the energy used onboard the ship, implying their collection and reporting by the ship is mandatory. Indeed, FuelEU Annex I also elaborates on the specific information that EDNs should – at least – contain.

The practicalities of a company's procedure to gather EDNs is expected to be variable. When assessing the procedure as stated in the monitoring plan, verifiers might consider the following elements as important to ensure the company will be in a position to submit all EDNs during the verification period:

- Does the procedure define a responsible person/department onboard the ship or within the company for ensuring EDNs are received after the port call?
- Does the procedure include any element of ensuring data quality? For example a cross-check by the chief engineer or electrician that the delivered electricity on the EDN matches what was actually received onboard? (Similar to cross-checks which may already take place for BDNs).

Note: During the first FuelEU reporting and verification periods, there have been challenges for companies to gather EDNs. In some cases, ships might not receive an EDN at all while in others the retrieved documentation might not contain all of the information required for EDNs as per FuelEU Annex I. Guidance on how verifiers might approach incomplete EDNs is addressed in section 3.2.

2.1.4 Methods to be used to estimate energy consumption from OPS in case of data gaps

According to the FuelEU monitoring plan template, Table D.2 (see below) should be completed where OPS consumption is applicable for ships. It is therefore recommended that verifiers require this table to be completed for any ship with equipment to allow connection to OPS (and for which Table B.4 of the template should already be completed).

Table 5 – FuelEU monitoring plan template – **Table D.2**

Methods to be used to estimate energy consumption from OPS and ZET in the event of data gaps

Title of method	Method to be used to estimate energy consumption
Back-up monitoring method	
Formulae used	
Description of method to estimate energy consumption	
Name of person or position responsible for this method	
Data sources	

Location where records are kept	
Name of IT system used (where applicable)	

It is recalled that, prior to the verification of a FuelEU report, shipping companies are – on the basis of article 11(b) of EU 2024/2027 – expected to provide the verifier the following information where data gaps occurred during the reporting period:

- the number of voyages for which data gaps occurred, and the circumstances and reasons for data gaps that apply;
- the estimation method for surrogate data applied;
- the amount of energy calculated based on surrogate data;

Therefore verifiers should, when assessing monitoring plans, consider whether a company's back-up method(s) to monitor energy consumption from OPS can be expected to provide surrogate data that they as a verifier can consider to be sufficiently reliable, credible, accurate and complete.

Recommended methods to be used to estimate energy consumption from OPS are, with the following priority:

1. Onboard measurement of electricity
2. Calculation of a ship's average electrical demand at berth multiplied by hours at berth

The application of both of these back-up methods should be subject to confirmation that onboard auxiliary engines were not in operation during the ship's stay at berth. The latter could be confirmed, for example, by engine room logbook records on running hours or fuel consumption of the auxiliary engines during the stay at berth.

Onboard measurement of electricity

Electricity consumption onboard the ship could be measured – for example – via onboard metering. Metering in the context of OPS has 2 objectives: to measure the electricity supplied for contractual reasons, but also to measure the quality parameters and electricity quality control which is a central functional requirement for SSE²⁵. As such, it is likely that OPS connection points installed on ships include a metering unit from which the electricity supplied to the ship's electrical grid can be monitored (for instance through a control panel on the engine control room/switchboard).

When accepting a supplementary or back-up monitoring procedure which relies on such onboard metering of OPS electricity received, verifiers might consider the following:

- Was the meter installed together with the on-board OPS equipment which adheres to an international technical specification (see section 0) with clear system integration evidenced by documentation?
- Does the meter have a type approval certificate (or similar) demonstrating that it is suitable for use onboard ships?
- Are there records available showing the accuracy of the meter as compared to an EDN?
- Have the verification team witnessed the functioning of the meter or seen an explanation during site visits?

²⁵ [EMSA Guidance on SSE Part 1](#) Section 3.1.1

- Is there documentation to show the meter has been installed in accordance with classification society rules?

Calculation of ship's average electrical demand at berth multiplied by hours at berth

Verifiers might also accept a procedure where, in case of data gaps, the shipping company estimates OPS energy consumption as the ship's average electrical demand (kW or MW) multiplied by hours at berth. In such a procedure, the shipping company might be required to state the ship's average electrical demand in Table D.2 of the monitoring plan and explain how the average was determined (for assessment by the verifier).

Verifiers might assess the reliability of the stated average electrical demand for the ship by comparing the value to the following example sources:

- The ship's established total electrical power demand given in Table B.6 of the FuelEU monitoring plan (for which the average can be expected to be moderately less)
- Historical data for the ship or a sister ship
- Literature estimates of average power demand at berth per ship type (see Table 6 as an example).

Table 6 Average power demand at berth per ship type

Ship Type	GT Sizes	Power Demand Average (MW)
Oil tankers	<5,000	4
	5,000-10,000	6
	>10,000	8
Chemical/product tankers	<5,000	6
	5,000-10,000	9
	>10,000	10
Gas tankers	<5,000	5
	>5,000	9
Bulk carriers	<50,000	0.5
	>50,000	2
General cargo	<25,000	1.5
	>25,000	3
Containerships	<10,000	1.5
	10,000-50,000	2
	>50,000	4
RO-Pax	<20,000	2

Cruise ships	>20,000	5
	<50,000	4
	50,000-100,000	9
	>100,000	18

Source: EMSA Guidance on SSE to Port Authorities and Administrations, Part 2 Table 7.7 (“OPS typical requirements and standards for different ship types”).

When assessing the reliability of a shipping company’s procedure to estimate OPS consumption in the event of data gaps, the following considerations may also be relevant:

- Is the ship engaged in liner shipping with scheduled, consistent port calls? If yes, the ship’s energy consumption from OPS could be expected to be fairly consistent when calling the same specific port throughout a reporting period. As a result, some data gaps (e.g. the occasional non-availability of an EDN) might be considered less of a verification risk if the ship would have data on OPS energy consumption from other stays at berth at that port.
- Conversely, if the ship is engaged in tramp shipping, the port and length of stay at berth may be more variable: at some berths the ship might use its own cranes for loading/unloading while in others not, etc. As a result, the energy consumed by OPS might be highly variable from port to port so data gaps (e.g. the non-availability of an EDN) might be considered a higher verification risk.

2.2 Time spent at berth

FuelEU requires ships to establish procedures to monitor the time they spend at berth and anchorage and report on this in the FuelEU report. Compared to the MRV, this is a new monitoring and reporting parameter, expanding on “time spent at sea”. Table C.3.4 of the FuelEU monitoring plan template is shown below. There is also a similar table concerning the back-up method the company intends to use to monitor time in the event of data gaps.

Table 7 FuelEU monitoring plan template - **Table C.3.4**

Procedures for determining and recording the time spent at sea and at berth

Title of procedure	Determining and recording the time spent at sea from berth of port of departure to berth of the port of arrival, and time spent at berth (quayside and anchorage)
Reference to existing procedure	
Version of existing procedure	
Description of procedures (including recording and managing port departure and arrival information) (a brief description of the procedure can be provided if already existing outside the monitoring plan)	
Name of person or position responsible for this procedure	
Formulae and data sources	

Location where records are kept	
Name of IT system used (where applicable)	

More information on the above procedures, as well as the applicable definitions is given below:

Time at berth:

“Ship at berth” means, according to FuelEU Article 3(15), (which itself relies on Article 3, point (n) of the [MRV Regulation](#)):

“a ship which is securely moored or anchored in a port falling under the jurisdiction of a Member State while it is loading, unloading or hotelling, including the time spent when not engaged in cargo operations”²⁶

It follows that the time that should be monitored under this procedure is the time that is elapsed between a ship’s arrival at a berth and departure from that berth on any voyage which is in scope of FuelEU. Note however that this parameter includes only time spent at berth in ports within the EEA. This is because the Regulation does not apply to the energy used by ships during a stay at a port under the jurisdiction of a third country²⁷ (the exception being neighbouring container transshipment ports²⁸).

It is expected that, for most ships - and subject to each verifier’s assessment - procedures to monitor time spent at berth might not differ substantially from procedures that are already in place for the EU MRV to monitor time spent at sea. According to guidance for the MRV, time spent at sea might be monitored based on deck logbook entries for the departure from the last berth in a port and arrival at the first berth in the next port, possibly supplemented by GPS indications or daily noon reports²⁹. Verifiers might therefore accept companies to take a similar approach for monitoring time at berth, for example: logbook entries for the time elapsed between a ship’s arrival and departure from a berth.

When assessing the reliability of such a procedure for FuelEU, verifiers might – for example – request the company to provide:

- A copy of the ship or company’s internal SMS procedure for how arrival at/departure from berth should be recorded. Verifiers might, for instance, consider it important to check that the procedure is sufficiently clear and well communicated to all bridge officers, including during the handover of shifts or crew transfer periods.
- A copy of existing logbook entries from the ship, to see evidence of the proposed procedure for recording time at berth working in practice. Logbooks could be examined remotely or as part of a site visit.

²⁶ According to FuelEU Article 3(15) which relies on Article 3, point (n) of the [MRV Regulation](#)

²⁷ See FuelEU Guidance document for shipping companies, p21. According to FuelEU article 2(a), a ship’s energy used during their stay within a port of call only applies to those ports under the jurisdiction of a Member State. While half of the energy used by ships on voyages to/from third countries is in scope, voyages are by definition movements which originate from or terminate in a port of call. The energy at berth in the third country is not in scope. Note: The energy used by ships in a port of call located in an outermost region under the jurisdiction of the Member State can be exempted from the scope of Regulation on the basis of Article 2(4). However the time the ship spends at berth at that port of call is still expected to be monitored and reported.

²⁸ For which stops are excluded from the definition of port of call under article 3(11). Time at berth in these ports must simply be considered as time spent at sea.

²⁹ See section 7.4 page 82 of the [EU ETS and MRV Maritime General guidance for shipping companies](#)

- Evidence of supplementary procedures in place by the company or ship to ensure data quality. An example of the latter could be, for instance, a digital logbook set-up which triggers a reminder to the crew/company if the ship has not recorded an entry for the time of arrival/departure at berth within some hours of ETA, etc.

The time a ship spends at berth is, for the purposes of FuelEU, most critical for container and passenger ships. This is because these ships face penalties for non-compliance to their obligations to connect to OPS in accordance with Article 6 and the magnitude of the penalty is based, in part, on the number of hours spent in non-compliance at berth.

As such, due to the more stringent OPS regulations for passenger and container ships, verifiers might consider there to be a higher verification risk in the event of time at berth not being correctly or consistently recorded. On the other hand, accurate recording of time at berth is important for all ship types. As well as being a mandatory reporting parameter, time at berth can be an aid to verifiers when corroborating the accuracy and reliability of other reporting parameters.

Time at anchorage:

The reason for recording time at anchorage in the FuelEU context is due to the possible requirement for ships to also connect to OPS while at anchorage on the basis of Article 6(11), subject to decisions by Member States.

“Ship at anchorage” means, according to Article 3(16) of FuelEU:

“a ship at berth which is not moored at the quayside”

with the definition of “ship at berth” being in accordance with Article 3(n) of Regulation (EU) 2015/757:

“‘ship at berth’ means a ship which is securely moored or anchored in a port falling under the jurisdiction of a Member State while it is loading, unloading or hotelling, including the time spent when not engaged in cargo operations”

This means that, in the context of FuelEU, time at anchorage to be reported as part of the time at berth is defined as the time that a ship is anchored during a port call.

Time at anchorage as part of a voyage³⁰ is to be excluded here and can be reported separately in FuelEU Database.

The verification of information on time at anchorage might be verified by methods such as, for example, deck-log book entries, AIS data, communication with port authorities or, in the case of cargo transferred at anchorage, details about the loading/discharge periods.

2.3 Wind Assisted Propulsion Systems

Wind Assisted Propulsion systems (WAPS) need to be described in a ship's FuelEU monitoring plan in Table B.7 as shown below.

³⁰ To recall, FuelEU defines a voyage as “any movement of a ship that originates from or terminates in a port of call”. A “port of call” is where ships stop to load or unload cargo or to embark or disembark passengers. See definitions in Article 3.

Table 8 FuelEU monitoring plan template - **Table B.7***Table B.7.*

Description of Wind Assisted Propulsion systems (WAPS) (when applicable)

WAPS reference no.	WAPS (name, type)	WAPS installation date	Technical description of WAPS	Pwind	Pprop

The installation of WAPS can have a major positive impact on a ship's compliance to the FuelEU GHG intensity limits, earning the ship up to a 5% reduction on its annual GHG intensity via the f_{wind} reward factor as specified in Annex I of the Regulation (see equation 1 below).

Equation 1: Methodology to calculate the GHG intensity of the energy used onboard by a ship, including f_{wind} reward factor

GHG intensity $\left[\frac{gCO_2eq}{MJ}\right] = f_{wind} \times (WtT + TtW)$ Equation (1)	
WtT	$\frac{\sum_i^n fuel M_i \times CO_{2eq, WtT, i} \times LCV_i + \sum_k^c E_k \times CO_{2eq, electricity, k}}{\sum_i^n fuel M_i \times LCV_i \times RWD_i + \sum_k^c E_k}$
TtW	$\frac{\sum_i^n fuel \sum_j^m engine M_{i,j} \times \left[\left(1 - \frac{1}{100} C_{slip, j}\right) \times (CO_{2eq, TtW, i, j}) + \left(\frac{1}{100} C_{slip, j} \times CO_{2eq, TtW, slip, i, j}\right) \right]}{\sum_i^n fuel M_i \times LCV_i \times RWD_i + \sum_k^c E_k}$
f_{wind}	Reward factor for wind-assisted propulsion

The applicable f_{wind} reward factor is presently based on the theoretical propulsion support of the installed WAPS (being a function of Pwind / Pprop) and not the actual energy generated or saved by the use of the WAPS³¹. This increases the importance of having reliable information about the technical specifications of the WAPS in the monitoring plan since the f_{wind} reward factor is determined solely as a function of the “ P_{wind} / P_{prop} ” ratio with these values derived from technical certificates (primarily the EEDI/EEXI technical file – as elaborated on below).

As such, in accordance with Article 5(e) of Implementing Regulation 2024/2027 on verification activities for FuelEU, verifiers have an obligation to ensure that WAPS – and indeed other sources of energy and fuel – are accurately and completely described in the FuelEU monitoring plan.

Confirming a WAPS is installed:

³¹ This may change following a review of the Regulation, which is due by the end of 2027. In particular, Article 30(2)(h) states that the review should include the possibility to include (actual) energy provided by wind in the calculation of the ship's GHG intensity, subject to the availability of a verifiable method for monitoring and accounting of wind propulsion energy.

FuelEU Article 3(9) defines WAPS as “propulsion whether partial or full, of a ship by wind energy harnessed by means of wind-assistance propulsion systems such as, inter alia, rotor sails, kites, hard or rigid sails, soft sails, suction wings or turbines”.

FuelEU therefore takes a broad definition of the technology and does not specifically exclude any kind of WAPS. However, note that there may be certain WAPS that do not qualify for the f_{wind} reward if the power of the WAPS is low compared to the power of the ship’s main engines. It is nonetheless expected that all WAPS, regardless of their power, are recorded in the monitoring plan under Table B.7 since they can still be used to support propulsion and reduce energy in scope of FuelEU.

Verifiers are expected to rely primarily on a ship’s EEDI or EEXI Technical File to confirm the WAPS is installed onboard and to corroborate information about the WAPS as provided on the FuelEU monitoring plan. Other, additional, sources that verifiers might choose to corroborate the accuracy of information provided by the company about the WAPS could include, for example:

- Site visits
- Classification society certificates – such as evidence of compliance to rules of a WAPS notation, where applicable.
- Ship drawings showing the WAPS: such as the general arrangement plan, etc. Drawings should be stamped as approved by the Flag administration / RO or, where applicable for class requirements, the classification society.

The importance of the EEDI/EEXI Technical File as the primary document for the purposes of the assessment of the FuelEU monitoring plan for WAPS systems is stated further below.

WAPS technical standards:

FuelEU does not itself prescribe any standards or technical specifications that a WAPS must adhere to.

The use of any WAPS is therefore permitted under FuelEU provided the installation is reflected in the ship’s EEDI/EEXI Technical File.

Confirming the P_{Wind} / P_{Prop} values:

The reward factor for wind-assisted propulsion under FuelEU depends on the ratio between the effective wind power (P_{Wind}) and the installed propulsion power of the ship (P_{Prop}). The higher the power of the installed WAPS compared to the total main engine propulsion power of the ship, the higher the WAPS reward factor (to a maximum of 0.95, which corresponds to a 5% reduction on the ship’s GHG intensity according to FuelEU Annex I).

P_{Wind} and P_{Prop} must both be stated in table B.7 of the FuelEU monitoring plan. FuelEU Annex I requires that the source of each of these values derives from the ship’s EEDI / EEXI technical file which contains the information necessary for the calculation of a ship’s EEDI and EEXI³².

³² According to MARPOL Annex VI Regulation 23 (Attained EEXI) and 24 (Attained EEDI). The technical file is also on the basis from which a ship’s EEDI/EEXI is verified by the Flag Administration or RO

The FuelEU verifier is not expected to re-verify a ship's EEDI/EEXI Technical File but rather rely on its accuracy as prepared for the ship and accepted by the corresponding Flag Administration/RO. A ship carries a copy of its EEDI/EEXI technical file onboard since the International Energy Efficiency Certificate (IEE Certificate) requires that the technical file always accompanies it. The EEDI/EEXI technical file should be clearly marked as accepted by the corresponding Flag Administration/RO and, to ensure the latest version of it is used to source P_{Wind} and P_{Prop} values, the identification/verification number of the technical file should match that which is referenced on the ship's IEE certificate³³.

Note that for ships not subject to the EEDI/EEXI (such as diesel electric ships), the value of P_{Wind} must still be verified by a recognised body (such as a classification society) by following the EEDI/EEXI guidance in MEPC.1/Circ.896. This is to ensure a standardised approach to WAPS is followed for all ships subject to FuelEU

P_{Wind} :

P_{Wind} is the available effective power delivered by a WAPS in kW (see FuelEU Annex I and [MEPC.1/Circ.896](#)) derived by multiplying the effective power (P_{eff}) with the availability factor (f_{eff}):

$$P_{Wind} = P_{eff} * f_{eff}$$

[MEPC.1/Circ.896](#) provides details on the exact calculation of P_{Wind} (see 2.3.1).

The respective values required to calculate P_{Wind} should be available within the ship's EEDI/EEXI Technical File under section 5 ("description of energy saving equipment"). This is on the basis of the Technical File sample for both frameworks, available in Appendix 1 of [MEPC.365\(79\)](#) and the Appendix of [MEPC 351\(78\)](#).

In cases where individual f_{eff} and P_{eff} values are not available directly within the EEDI/EEXI Technical File, the FuelEU verifier may ask the shipping company to share documentation from the Flag Administration or RO from the period when the EEDI/EEXI calculation and verification was carried out.

P_{Prop} :

According to FuelEU Annex I, P_{Prop} is the propulsion power of the ship and corresponds to the power of the main engines (P_{ME}) as defined in [MEPC.365\(79\)](#) and [MEPC.350\(78\)](#).

P_{ME} is given in section 6.2 of a ship's EEDI/EEXI Technical File, according to the samples given in Appendix 1 of [MEPC.365\(79\)](#) and the Appendix of [MEPC 351\(78\)](#), - see the examples below.

Table 9 P_{ME} in Sample of EEDI Technical File (Appendix 1 of MEPC.365(79))

³³ According to MARPOL Annex VI Appendix VIII (on the Form of International Energy Efficiency Certificate) the identification/verification number on an EEDI/EEXI technical file is given in section 8.1.1 and 9.1.1 of the IEE certificate

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6 Calculated value of attained EEDI

6.1 Basic data

Type of ship	Capacity DWT	Speed V_{ref} (knots)
Bulk Carrier	150,000	14.25

6.2 Main engine

MCR_{ME} (kW)	Shaft gen.	P_{ME} (kW)	Type of fuel	C_{FME}	SFC_{ME} (g/kWh)
15,000	N/A	11,250	Diesel Oil	3.206	165.0

Table 10 P_{ME} in sample of EEXI Technical File (Appendix of MEPC 351(78))

6 Calculated value of attained EEXI

6.1 Basic data

Type of ship	Capacity DWT	Speed V_{ref} (knots)
Bulk carrier	150,000	13.20

6.2 Main engine

MCR_{ME} (kW)	$MCR_{ME,lim}$ (kW)	P_{ME} (kW)	Type of fuel	C_{FME}	SFC_{ME} (g/kWh)
15,000	9,940	8,250	Diesel oil	3.206	166.5

For most ship types, P_{ME} is in practice 75% of the rated installed power (MCR) for each main engine³⁴, however this can differ for LNG carriers depending on the method of propulsion.

Note also that for the EEXI, certain ships may also use Engine Power Limitation as a compliance method. In those cases, P_{ME} is – according to [MEPC.350\(78\)](#) – 83% of the limited installed power (MCR_{lim}) or 75% of the original installed power (MCR), **whichever is lower**. Accordingly, the lower value should already be stated as P_{ME} in the Technical File and is the value that should be used for P_{Prop} in the context of the FuelEU monitoring plan.

P_{prop} when shaft motors installed

Importantly, where shaft motor(s) are installed onboard the ship, P_{ME} is - according to FuelEU Annex 1 - calculated in a different way:

$$P_{Prop} = P_{ME} + P_{PTI(i),shaft}.$$

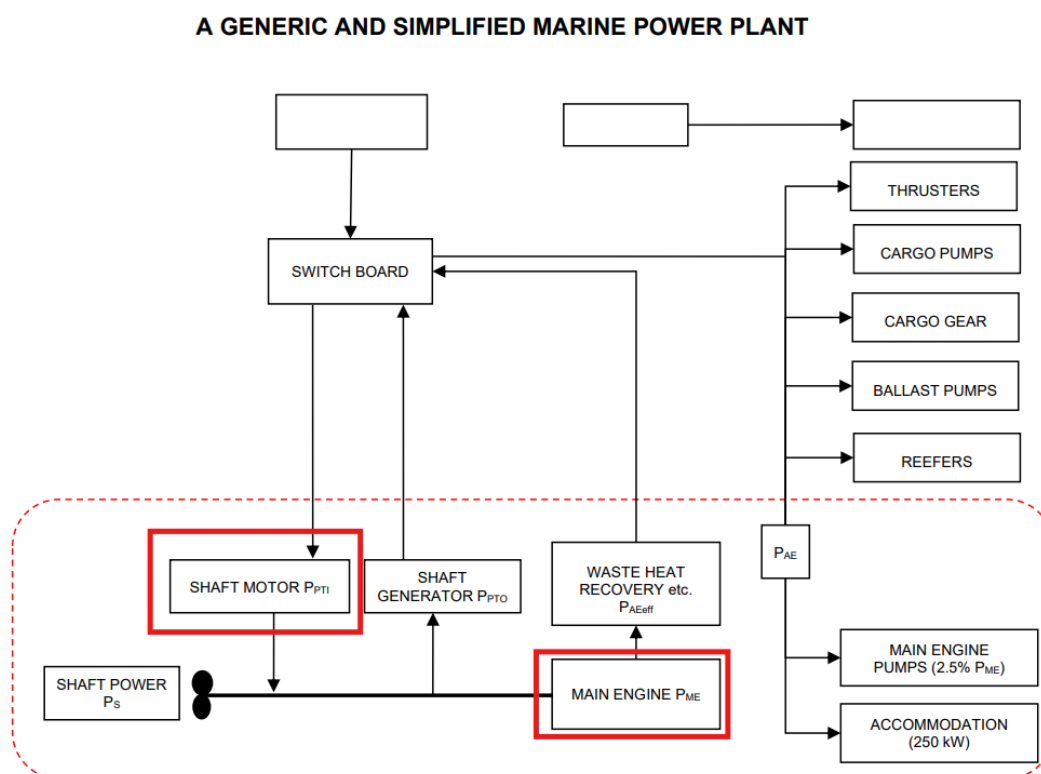
³⁴ According to MEPC 364(79). The value of MCR specified on the EIAPP certificate should be used for calculation. If the main engines are not required to have an EIAPP certificate, the MCR on the nameplate should be used.

$P_{PTI(i),shaft}$ refers to shaft *motors* for Power Take In (PTI) (where a motor is installed to support the driving of the crank/propeller shaft).

A schematic of the components of P_{Prop} (P_{ME} and P_{PTI}) is given in Figure 1 below from the corresponding EEDI guidelines.

Note: FuelEU does not specify P_{Prop} to be adjusted in case of shaft *generators* being installed (Power Take Off, or PTO). FuelEU only requires the adjustment for Power Take In (PTI)³⁵

Figure 1 Appendix 1 of MEPC 364(79) showing P_{ME} and P_{PTI}



Confirmation about whether a ship has P_{PTI} installed may be found in section 3 of the EEDI / EEXI Technical File (“Overview of propulsion system and electric power supply system”³⁶). The formula for P_{PTI} can be found in section 2.2.5.3 of [MEPC 364\(79\)](#). The P_{PTI} value should be available within section 5 of the EEDI/EEXI Technical File.

Accordingly, where ships have (Power Take In) shaft motors installed, the verifier should ensure that the value “ $P_{PTI(i),shaft}$ ” has been added to the value for P_{ME} for determining P_{Prop} in the FuelEU monitoring plan.

The application of f_{wind} in the verification of a FuelEU report

When it comes to the verification of a FuelEU report, verifiers are responsible to ensure the correct application of the f_{Wind} reward factor in Annex I of FuelEU (used to establish the GHG intensity of the ship in the reporting period).

The reward factor is determined as follows in Annex I of FuelEU:

³⁵ This is because according to the EEDI guidelines ([MEPC.364\(79\)](#)) P_{ME} would already be reduced to account for shaft generators when installed and this value for P_{ME} reflected on the ship’s Technical File.

³⁶ on the basis of the samples of the EEDI/EEXI Technical File in Appendix 1 of [MEPC.365\(79\)](#) and Appendix of [MEPC 351\(78\)](#).

Reward factor for wind-assisted propulsion – WIND (f _{wind})	Value of P_{Wind} / P_{Prop}
1 (thus no reward)	< 0.05
0,99	$0.05 \leq P_{Wind} / P_{Prop} < 0.10$
0,97	$0.10 \leq P_{Wind} / P_{Prop} < 0.15$
0,95 (maximum reward)	≥ 0.15

All P_{Wind}/P_{Prop} calculations should be rounded upwards at the 5th decimal place.

To take an example, a ship with a P_{Wind} of 1,000 kW and a P_{Prop} of 19,780 kW would have a P_{Wind}/P_{Prop} ratio of 0.05056, which earns a reward factor of 0.99. This corresponds to a 1% reduction that is applied to the ship's annual GHG intensity.

Note: [ESSF SAPS WS1](#) report has provided guidance on mid-year installations of WAPS. In line with this guidance, verifiers should consider a WAPS installed whenever the EEDI/EEXI technical file has been updated and verified to include the calculation of wind-assisted propulsion contribution. In the case of mid-year installation, the same reward factor is applied for the FuelEU reporting entire year.

Similarly, if a ship would undergo changes which affect its P_{Wind} / P_{Prop} ratio, (such as a change to the WAPS or a replacement/de-rating of the main engines) the verifier should consider these revisions complete when the EEDI/EEXI technical file has been updated and verified to include the calculation of wind-assisted propulsion contribution. In this case, the more recent reward factor is applied for the entire year.

2.4 Consumption of substitute sources of energy (or ZET)

“Substitute sources of energy” are defined in FuelEU Article 3(8) as meaning:

“renewable energy generated on board or electricity supplied from OPS”

FuelEU Article 8(3)(g) requires a ship's FuelEU monitoring plan to include a description of the procedures for monitoring energy provided by substitute sources or a zero-emission technology. FuelEU Article 15(1)(f) also requires ships to monitor and record the amount of each type of substitute source of energy consumed **at berth and at sea**. Subsequently, the FuelEU report is required to state the amount of each type of substitute source of energy consumed by the ship **at sea** (Part E(h) of the report).

It is therefore important that ships have a procedure in place to accurately monitor consumption of energy from “substitute sources of energy”.

Based on the definition of substitute sources of energy, examples might include the following energy sources, in addition to electricity from OPS:

- Fuel cells using the chemical energy from renewable sources
- Energy storage systems / batteries which have their power supplied from renewable sources.
- On-board power generation from wind and solar energy

While there is no dedicated table in the FuelEU monitoring plan template which specifically covers substitute sources of energy³⁷, it is recommended that a procedure for monitoring energy provided by substitute sources is provided in the table for ZETs (Table C.1.7.). This recommendation is in line with the current set-up of the FuelEU Database environment which allows the creation/edit of a FuelEU monitoring plan.

However, strictly speaking, while technologies used as substitute sources of energy may overlap with ZETs in practice (for example, fuel cells or energy storage systems charged from renewable generation), they have different definitions³⁸, rules³⁹ and purposes⁴⁰ under the Regulation. Most significantly, as of the date of publication of this document, secondary legislation setting out the detailed acceptance criteria⁴¹ for technologies to be considered as fulfilling the general requirements for zero-emission technologies in the context of FuelEU Article 6 for energy requirements at berth (instead of OPS) is still to be adopted.

The following information is recommended to be included in the procedure for monitoring energy provided by substitute sources:

- Installation date
- Rated power of the energy source (in kW)
- Source of renewable energy
- The means of monitoring energy consumption from this energy source
- Description of onboard energy distribution: such as whether the power is distributed directly to the ship grid or to batteries, or only to certain consumers.
- The phase of ship operation where the substitute source of energy is intended to be used (e.g. open sea, port areas, etc)

In order to corroborate the accuracy of the above information, verifiers might ask companies to provide copies of the following example documentation:

- Classification society certificates / reports as confirmation the energy source adheres to all relevant class (or statutory) rules
- Ship drawings approved by the flag/RO/classification society: such as electrical distribution diagrams

Importantly, article 13(3)(d) of Implementing Regulation (EU) 2024/2027 notes that verifiers should consider the amount of energy obtained/derived from substitute sources of energy to represent an area of higher verification risk. As such, verifiers are recommended to ensure that for each source of substitute energy onboard there is a clear procedure in the monitoring plan for how the energy will be monitored. **Due to the higher verification risk substitute sources of energy are also recommended to be a point of attention during site visits.**

³⁷ The FuelEU monitoring plan template provides Table C.1.6. and Table C.1.7 for ZETs. However neither table fully captures the broader category of “substitute sources of energy” because the above are technologies intended for energy consumption at berth in the context of the zero-emission at berth obligations under Article 6 of FuelEU. Indeed, the FuelEU report requires that only the amount of energy consumed from ZETs at berth is reported. In contrast, substitute sources of energy may also be used at sea, and such usage must also be monitored and reported.

³⁸ ‘zero-emission technology’ means a technology that, when used to provide energy, does not result in the release of the following greenhouse gases and air pollutants into the atmosphere by ships: carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), sulphur oxides (SO_x), nitrogen oxides (NO_x) and particulate matter (PM);

³⁹ The use of ZETs in the first reporting period was not possible to a lack to defined acceptance criteria

⁴⁰ See footnote 39. A ZET is in principle used only **at berth** as a compliance measure to FuelEU article 6. Whereas substitute sources of energy are, in the context of FuelEU, intended to be used also at sea.

⁴¹ including the definition of system boundaries and certification requirements

2.5 Monitoring fuel consumption, time and distance travelled in ice conditions

Compared to the EU MRV/ETS framework, FuelEU contains an additional derogation related to ice-classed ships – namely a deduction of additional energy used when sailing in ice conditions. The other FuelEU derogation related to ice (due to the technical characteristics of ships with an ice class IA, IA Super or equivalent) also exists for the EU ETS framework and so is not discussed in this specific guidance for FuelEU verification⁴².

In order to ensure the correct application of the sailing in ice conditions derogation, there are mandatory procedures which need to be reflected in the FuelEU monitoring plan for any ship intending to make use of the derogation:

- Procedures for determining and recording the date, time and position when entering and leaving the ice conditions, the distance travelled when sailing in ice conditions (Table C.3.3, as well as Table D.4. concerning methods to treat data gaps)
- Procedures for monitoring fuel consumption (of each fuel consumer) when sailing in ice conditions (Table C.1.8)

It is recalled that the derogation for sailing through ice is only applicable for ships with ice classes IA Super, IA, IB and IC (or equivalent⁴³). As such, when reviewing the procedure during the assessment of the monitoring plan, the first step could be to check the ship has the correct ice class to be eligible for the derogation in the first place. The ship's ice class should already be stated in Table B.1 of the FuelEU monitoring plan, with this entry mandatory for any ship seeking to use either of the FuelEU ice-class related derogations. The accuracy of the ship's stated ice-class could be confirmed by valid class or statutory certificates such as:

- Polar Ship Certificate⁴⁴
- Certificate of classification, where additional class notations (such as for sailing in ice) are normally stated.
- Cargo Ship Safety Construction Certificate (in accordance with SOLAS Chapter II)
- Passenger Ship Safety Certificate

When assessing the adequacy of a ship's proposed procedures for monitoring fuel consumption when sailing in ice conditions (Table C.1.8), and time and distance through ice (Table C.3.3), verifiers should consider whether the procedure is likely to result in reliable data to enable the application of the FuelEU derogation.

In the following subsections, the general meaning of "sailing in ice conditions" under FuelEU is first recalled. Thereafter, some considerations are given for each of the procedures above in terms of how the verifier might assess their adequacy.

The meaning of "sailing in ice conditions"

⁴² For general background on both ice-derogations for FuelEU, see section 4.6 of the [Guidance on the FuelEU maritime regulation](#) for shipping companies.

⁴³ HELCOM Recommendation 25/7 describes the relationship different ice classes, which may be part of a ship's classification society notation. Note that the names of class notations may change, so companies and verifiers are advised to check with the classification society that issued the ice notation if the equivalency of it to IA Super, IA, IB and IC is uncertain. Where the verifier is not the Classification Society, the ice class notation issued should be accepted as stated on the ship certificate.

⁴⁴ According to the form of the certificate given in Appendix 1 of the [Polar Code](#), Ice class is stated in point 2 of the certificate

For the purposes of FuelEU, “sailing in ice conditions” means – according to FuelEU Article 3(23) - **the sailing by an ice-class ship in a sea area within the “ice edge”**.

“Ice edge” means – according to FuelEU Article 3(23) - **the demarcation at any given time between the open sea and sea ice of any kind**, whether fast⁴⁵ or drifting, as set out in paragraph 4.4.8 of the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#). The referenced World Meteorological Organisation paragraph clarifies also that ice edge may be termed compacted (meaning a close-clear-cut ice edge⁴⁶) or diffuse (meaning a poorly defined ice edge limiting an area of dispersed ice; usually on the leeward side of an area of drift ice⁴⁷).

For the sake of completeness, “sea ice” itself is any form of ice found at sea which has originated from the freezing of sea water⁴⁸. “Open water” is in contrast “a large area of freely navigable water in which sea ice is present in concentrations less than 1/10. No ice of land origin is present.”⁴⁹

It follows, then, that – under FuelEU – the definition of “sailing in ice conditions” is broad, including the sailing through ice of any kind.

Procedures for determining and recording the date, time and position when entering and leaving the ice conditions, the distance travelled when sailing in ice conditions

FuelEU article 8(3)(o) specifies that, where the sailing in ice derogation is used, the FuelEU monitoring plan shall include a “verifiable procedure” for monitoring the distance travelled for the whole voyage as well as the distance travelled when sailing in ice conditions.

This is because it is expected that sailing through ice would in many cases be applicable only to part of a ship’s voyage, so delineating clearly when a voyage is through ice or in open seas is highly important. Without a clear delineation, determining the additional fuel consumption specifically through ice (which is what is needed to apply the derogation) might result in inaccuracies or incompleteness.

In order to assess whether a procedure is sufficiently “verifiable”, verifiers might consider the following elements:

- Does the procedure clearly indicate how the sailing through ice will be documented, for instance requiring entry in a deck logbook when sailing in ice starts, followed by an entry when sailing in ice ends? To assess such a procedure, the verifier might for example ask for copies of records that have already been made by the crew in these cases, or a copy of SMS or internal procedures showing the procedure is communicated to all navigational officers to ensure consistent logbook recording.
- Are there steps in place to ensure data quality, such as systems which do not only rely on manual logbook entries? An example could be software programmed to create automatic log registries when entering ice navigation.

⁴⁵ Meaning ice formed along the coast and which may extend a few metres or several hundred kilometres from the coast. See full description in paragraph 1.1.1 of the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#)

⁴⁶ See paragraph 4.4.8.1 of the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#)

⁴⁷ See paragraph 4.4.8.2 of the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#)

⁴⁸ See paragraph 1.1 the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#)

⁴⁹ See paragraph 4.2.6 of the [World Meteorological Organisation Sea-Ice Nomenclature, March 2014](#)

- Has the ship previously monitored and reported distance through ice on a voluntary basis for the MRV? If yes, the soundness of the procedure could be assessed by the verifier by consulting any recommendations for improvement from previous MRV verification cycles.

Procedures for monitoring fuel consumption (of each fuel consumer) when sailing in ice conditions:

In order to assess a company's procedure for monitoring fuel consumption when sailing in ice conditions, verifiers might consider the following elements:

- Are all relevant consumer types onboard addressed in the procedure (main engines, boilers, auxiliary engines)?
- If particular equipment is needed for this monitoring (such as flow meters) are they installed for all relevant consumers?
- Is there a procedure in place onboard the ship to ensure fuel consumption measuring equipment is calibrated?
- Does the company provide information on the level of accuracy of the fuel consumer monitoring equipment and is this accuracy also valid for short events in the course of a single voyage such as intermittent sailing in ice, open water, ice, open water, and so on?

Note: While the regulatory requirements of the IMO DCS and Carbon Intensity Indicator are different to those of FuelEU, for general guidance related to methods for verification of sailing in ice conditions and associated fuel consumption readers might consult section 11.1 of IACS Recommendation 175⁵⁰ and MEPC.355(78)⁵¹. This being said, FuelEU and its secondary legislation remain the leading framework for this matter.

3 Verification of FuelEU reports

There are several areas of the FuelEU report template (Annex I ([\(EU\) 2024/2027](#))) which differ from a ship MRV report template (Annex II ([\(EU\) 2023/244](#))), due to the differences in the scope and policy goals of the Regulations.

The additional FuelEU report parameters are listed in column 1 of Table 11 below while columns 2 and 3 show the availability of existing guidance in other documents. Column 4 lists the corresponding section of this document where additional guidance can be found.

⁵⁰ [IACS Rec No. 175: SEEMP/CII Implementation Guidelines](#). Note: the most recent version should always be consulted.

⁵¹ Concerning the Interim Guidelines on correction factors and voyage adjustments for CII calculations

Guidance on verification for FuelEU Maritime Regulation

Table 11 Identified areas of FuelEU report verification which differ from verification of MRV ship emission reports

Item	(Partially) addressed in CLIMA GD1 or GD 3?	(Partially) addressed in FuelEU guidance for shipping companies?	Location of additional guidance in this document (section)
Items from FuelEU report ⁵² subject to verification not already part of MRV			
Time spent moored at the quayside or time spent at anchorage - Part D(d) of FuelEU report	No	Yes, chapter 6	3.1
Energy supplied to the ship via OPS Part E(a) and (e) of FuelEU report	No	Yes, chapter 6	3.2
Exceptions to OPS connection Part E(b) of FuelEU report	No	Yes, Chapter 6.1.2. Describes the ports where OPS connection obligation applies and the 7 exemptions which may apply.	N/A Guidance may be developed ahead of 2030
Aggregated amount of fuel consumed on voyages / while moored in port calls subject to a derogation on basis of article 2(4-6) - Part E(c) (d) of FuelEU report	The MRV/ETS exemptions are explained in 2.3.3 of GD1.	FuelEU exemptions/derogations are explained in 3.3.2.	N/A < see FuelEU guidance for shipping companies and the up to date list of exemptions applied by Member States
Amount of energy consumed at berth from ZETs - Part E(g) of FuelEU report	No	No	N/A Pending acceptance criteria for ZETs
Amount of each type of substitute source of energy consumed at sea - Part E(h) of FuelEU report	No	No	3.3
Reward factor for wind assisted propulsion - Part E(i) of FuelEU report	No	Yes, chapter 4.5.	2.3
Data for sailing in ice exemptions - Part F of FuelEU report	Yes, for the ETS derogation for ships with IA/IA Super or equivalent (technical characteristics).	Yes, chapter 3.6 [both FuelEU ice derogations explained in with examples]	3.4

⁵² See template in Annex I of Implementing Regulation (EU) 2024/2027

Verifier calculations to be made on basis of the FuelEU report (article 16(4) of the FuelEU Regulation)			
Annual GHG intensity as per Annex I	N/A	Yes, examples in chapter 4.2	See FuelEU guidance for shipping companies
Compliance balance as per Annex IV	N/A	Yes, examples in Chapter 10.2	See FuelEU guidance for shipping companies
Number of non-compliant port calls	N/A	No	N/A as non-compliant ports calls are a possibility only from 1 January 2030
Amount of the energy excluding energy from OPS	N/A	No.	No specific guidance given. It is a case of subtracting the verified value for Part E(e) from the ship's total energy consumed in scope
Amount of energy from RFNBOs	N/A	No.	No specific guidance given. Part E(d) ¹ requires the total amount of each type of fuel consumed within the scope of Article 2 of Regulation to be reported. The LCV of each RFNBO is also given in Annex II of the Regulation, or to be provided by the company via the PoS/PoC. The total amount of energy can be calculated based on the above verified values.

3.1 Time spent moored at the quayside and/or time spent at anchorage

Time spent moored at the quayside and/or time spent at anchorage has to be reported in Part D(d) of the FuelEU report.

Assessing the reliability of data reported for time spent at berth

When it comes to the verification of a FuelEU report, verifiers might consider the following elements when assessing the reliability of reported data for time spent at berth. These checks might be conducted either by means of sampling or – where the verifier considers there to be a higher verification risk – more systematic controls on the reported data.

- Cross checks of the ship's reported time at berth with external data sources, such historical AIS data. Experience from the MRV shows that in cases of good AIS coverage, the data appears to be a quite reliable external set of reference data - at least for "distance travelled" and "time spent at sea"⁵³.
- Consider the consistency of the ship's reported duration at berth with the ship's previous port calls in the same port: are there outliers without reasonable explanations?
- In the case of ships which connected to OPS during the time at berth: is the reported arrival/departure time consistent with the connection times stated on the Electricity Delivery Note?
- If a ship bunkered fuel during a stay at berth: does the reported time at berth correspond with time stamps on the Bunker Delivery Note?
- Are there pilotage, tug boat or mooring line records which can corroborate the ship's stated berth arrival/departure time?
- Request the company to share copies of the documentation exchanged between the port/terminal authority prior to arrival/departure to the berth or port. This could include reporting formalities toward port (National) Single Window Systems.
- Examine the consistency between the amount of fuel or energy that was reported to have been consumed at berth and the reported time at berth:
 - For ships which consumed OPS during the stay at berth: compare the time at berth and the ship's average⁵⁴ or total established electrical power demand at berth as per the kW value stated in table B.6. Is the reported time at berth plausible for the total amount of kWh claimed to have been consumed?
 - For ships which consumed fuel for electrical power at berth: check the specific fuel oil consumption stated for the ship's auxiliary engines as provided in Table B.3 of the FuelEU/MRV monitoring plan. If a typical power factor⁵⁵ and engine load is assumed, is the amount of fuel consumed in this period consistent with the reported time at berth?

⁵³ Section 8.7 [EU ETS and MRV Maritime Guidance for Accreditation and Verification](#)

⁵⁴ See section 2.1.4 of this document. The FuelEU monitoring plan template does not require that a ship's "average" electrical power demand at berth is stated in the monitoring plan. However this average might be stated in Table D.2 of the monitoring plan if it is used as a means to estimate energy from OPS in the event of data gaps.

⁵⁵ A power factor of 0.8 for ship auxiliary engines appears to be typical. See <https://www.emsa.europa.eu/electrification/sse.html>

3.2 Amount of electricity delivered to the ship via OPS

The amount of electricity delivered to the ship via OPS has to be reported in Part E(e) of the FuelEU report.

Assessing the reliability of reported electricity from OPS via EDNs

The primary verification activity for ensuring the reliability a ship's reported electricity consumption from OPS is expected to be a review of the consistency of the reported electricity with the corresponding Electricity Delivery Note (EDN). This expectation is because shipping companies are in any case required to share EDN's with the verifier on the basis of Article 11.2(d) of [Implementing Regulation \(EU\) 2024/2027](#) on verification activities. The later furthermore emphasises that the EDN provided to the verifier should be "complemented pursuant to Annex I" of FuelEU (meaning the EDN should contain information such as supplier, receiving ship, port, etc).

When assessing the consistency of the energy reported by the ship and the energy as stated on the EDN, verifiers may take a similar approach as they do when reviewing BDNs for fuel bunkering.

Elements might include the following:

- Does the date, time and port of call on the EDN match what the ship has reported?
- Is the stated amount of electricity plausible for the ship? A (conservative) rule of thumb could be multiplying the hours at berth with the kW value of the ship's "established total electrical power demand of the ship at berth" given in table B.6 of the monitoring plan.

Assessing the reliability of reported electricity from OPS in case of data gaps

Challenges across the industry have been noted during the first FuelEU reporting year with regard to the timely exchange of EDN's containing all the information set out in FuelEU Annex 1. Missing EDNs create data gaps, potentially requiring companies to use backup methods from Table D.2 of their FuelEU monitoring plan.

When assessing reported data reliability in such cases, verifiers might consider the following elements:

For all ships:

- FuelEU Article 6(8) requires ships to inform port authorities prior to entry about their intention to connect to OPS and also indicate the amount of power they expect to require during that port call. The verifier could ask the shipping company to provide this information and compare the expected and reported electricity consumption.
- Request companies to share (internal) shipboard documentation regarding the estimated power consumption they expected during the stay at berth. Such an estimate is recommended to be included in an OPS pre-connection checklist in line with the guidance in MSC.1/Circ.1675⁵⁶.
- Consult sources such as the [European Alternative Fuels Observatory OPS infrastructure map](#) to check whether OPS infrastructure exists at the port, including at the required output electrical capacity, for the ship type which has claimed the use of OPS. This could be in particular relevant in reporting years before 2030 as the infrastructure is still being rolled out.

⁵⁶ See section 5, point 9

- Check information directly from the port authority/terminal where a ship has claimed to receive energy from OPS: was there a connection point available at the berth where the ship claims to have used it? Does AIS tracking data show that the ship was berthed in a configuration that would allow an OPS connection? (check the description and position of the ship's OPS connection on table B.4 of the monitoring plan)
- If OPS consumption is estimated because of incomplete or insufficient EDNs / electricity supply declarations as opposed to fully absent EDNs, review the documentation that has been provided by the shipping company. It could be that part of the energy consumption is recorded. On this basis, the verifier could consider extrapolating the energy consumption for the ship's full time at berth.
- Consider that there is always a lag-period between the ship's arrival / departure from a berth and the connection/disconnection of the OPS cabling. If a shipping company claims to have received OPS for a duration the same or longer than their total stay at berth, this suggests a misstatement to be clarified.
- In cases where a different verifier assessed the ship's monitoring plan as satisfactory, check that ship's established total electrical power demand at berth has been calculated taking into consideration all available guidance. Then use that value and the stated hours at berth to conservatively assess whether the total kWh consumed is realistic.
- In the case of ships with cargo transfer cranes or pumps: the use of these at the berth would have a large impact on the electrical energy demand. Ask the company to provide evidence these were used, for instance the running hours of the machinery.
- Note, for ships also in scope of the IMO DCS, future/parallel IMO data fields may include total onshore power supplied in kWh⁵⁷ and may be useful as a consistency cross-check for total OPS consumption on an annual aggregated basis, but cannot replace FuelEU EDNs and FuelEU-specific evidence requirements.

For containerships and/or other ships with reefer cargo units:

The number of active reefer containers/trailers/cargo units onboard can have a large bearing on the electrical power demand of the ship.

A broad average for the power demand of a single refrigerated container is considered to be 3.6 kW per TEU (20 foot container) while a 40 foot container is around 7 kW (GDV, ongoing).

The number of active reefer containers onboard during the time of the OPS connection could be obtained by cargo manifests or other records provided by the shipping company. A verifier could examine the energy delivered by OPS to the ship for a port of call for which there are no data gaps and the number of active reefers that were onboard then as compared to the number of active reefers onboard for the OPS port call with data gaps.

Note: A dry-dock or repair facility is not considered to be a port of call according to Article 3(10) of the FuelEU Maritime Regulation. That being said, in some cases, energy consumed by a ship in dry dock or other facilities/stops outside of a port of call could still be in scope of FuelEU if they take place as a part of a wider voyage in scope of the Regulation (see section 3.3.3 of the FuelEU guidance for shipping

⁵⁷ MEPC.403(83) includes onshore power supplied as a data field in the MARPOL Annex VI context.

companies⁵⁸). As such, shore-electricity consumed by the ship during these specific cases can be considered in scope of the Regulation for which the amount of kW consumed by the ship should be confirmed via an EDN. If an EDN is not possible, the shore electricity consumed onboard the ship may otherwise be estimated in line with the above guidance to the satisfaction of the verifier. In this context, it is recalled that only electricity consumption onboard the ship would be in scope and must be distinct from shore electricity consumption related to the wider dry-docking maintenance or other event (such as shore cranes or other equipment in the case of a dry-docking) which would not be in scope.

3.3 Amount of energy from substitute sources consumed

The amount of substitute sources of energy consumed at sea must be reported in Part E (h) of the FuelEU report.

As stated before, article 13(3)(d) of Implementing Regulation (EU) 2024/2027 notes that verifiers should consider the amount of energy obtained/derived from substitute sources of energy to represent an area of higher verification risk. As such, in addition to appropriate monitoring, verifiers are advised to **ensure that the use of substitute sources of energy has been correctly allocated** to consumption at berth and consumption at sea, as well as ensuring **no double counting** of energy has occurred. With that purpose, this section outlines some ways in which companies may need to report the substitute sources of energy, and relevant documentation that verifiers could consult to ensure the veracity and accuracy of what has been submitted in the FuelEU report.

For OPS, the main method for deriving the amount of energy consumed is using EDNs or energy bills, as outlined above. Onboard metering may be used as an alternative in case of data gaps. For substitute sources of energy other than OPS, deriving the amount of energy consumed may be done using equipment technical information and onboard metering.

Assessing the accuracy of reporting between instantaneous energy consumption and charging batteries for later consumption

Some ships may charge onboard batteries at one point in time and then discharge these batteries later. This is currently not accounted for in the monitoring methods. However, in practice, reported consumption may be split between instantaneous energy consumption and charging the batteries for later use. Instantaneous consumption is reported when it takes place, but the energy from batteries is only reported upon discharging.

It is up to the verifiers to confirm that energy flows have appropriately split between instantaneous energy consumption and battery charging. This split must be based on the magnitude of the energy flows going to instantaneous consumption versus battery charging, as measured using the monitoring systems detailed in Section 2.4.

This typically occurs when companies charge their ship's batteries at berth, for use at sea. For those ships, the energy acquired at berth must be appropriately split between the at-berth consumption and the

⁵⁸ Available at https://transport.ec.europa.eu/transport-modes/maritime/decarbonising-maritime-transport-fueleu-maritime_en#secondary-legislation

at-sea consumption. At-berth energy consumption includes, for example, the hotel and cargo handling loads. At-sea energy consumption includes all energy used to charge a battery whose energy is deployed at sea.

For energy from OPS, shipping companies register this in the FuelEU Database as at-berth (Part E(e) of the FuelEU report) and at-sea OPS (Part E(h) of the FuelEU report), respectively, where at-sea consumption is the difference between the energy supplied from OPS and the at-berth use. This is detailed further in Section 2.4.

Verifiers must assess the accuracy of shipping company reports on instantaneous energy consumption versus charging batteries for later consumption. Verifiers must also ensure that all incoming energy is accounted for. This includes conversion/transmission losses, battery leakage, and battery discharge inefficiencies. Shipping companies are not to report these losses separately. Instead, they are to report them as part of the ship's consumption, depending on when the energy was consumed. For example, verifiers must ensure that at-sea battery discharge inefficiencies are considered under a ship's energy consumption for the relevant voyage. To verify the reliability of the reported data, verifiers may take the following example actions:

- Actions related to the delivery of electricity via OPS as established in Section 3.2
- Request ship information that would allow them to clearly establish how the batteries can be charged (e.g. relevant schematics), to establish whether they can be charged with sources other than OPS that may need to be treated differently⁵⁹
- Considering whether the reported electricity consumption at berth and at sea are realistic, for example, comparing with technical sheets for electric propulsion or the typical consumption at berth of similar ships
- Ensure that the sum of the OPS electricity used at berth and at sea is in line with the total supplied via OPS according to EDNs
- For ships equipped with electric propulsion and batteries, reporting zero-emissions voyages: assess whether it is realistic to cover the voyages with the battery capacity and propulsion output, and cross-check that EDNs match the expected charging needs.
- Where applicable: request software logs of energy management systems on ships that can indicate the timestamped charging and deployment of batteries, electric propulsion output, and other relevant information.

Assessing the accuracy of reported energy and the corresponding emissions factor

Where a ship uses a mix of energy sources (e.g. at sea usage of both OPS-charged batteries and diesel engines), shipping companies must monitor the energy provided by each source so that consumption can be allocated correctly, and verifiers must ensure this is reported correctly.

Shipping companies are not expected to face major technical hurdles in the monitoring of different inflows of electricity, provided accurate measurements are available at the OPS connection points and onboard generators. It is expected that modern ships can do highly accurate readings right at the high voltage shore connection switchboard (6.6–11 kilovolts). Older ships may be less accurate but can still

⁵⁹ Given the higher verification risk previously expressed, it is better if a site visit has been performed during a prior assessment of the monitoring plan, to ensure the monitoring plan and ship documentation match reality.

measure at current/voltage transformers and local meters. Nonetheless, as stated in Section 2.4, substitute sources of energy may present a higher verification risk, so it is advisable for verifiers to pay close attention to shipping companies' reports on allocating energy to the correct source.

For OPS-provided energy, companies may currently treat the WtT emissions factor as zero under the FuelEU Maritime Regulation. In addition to the methods described in Section 2.4, companies may report their OPS consumption using EDNs or OPS energy bills. Recall that when OPS-electricity is stored for later use, as described above, additional information beyond EDNs or energy bills is necessary to split the energy consumption between instantaneous consumption and battery charging. It is therefore important that verifiers ensure that reports contain all necessary information.

Electricity generated from fuel-based sources onboard is considered energy consumption from fuel and the applicable emissions factors are those set by the Regulation (Annex II). In this case, verifiers should ensure that electricity generated from fuel-based sources is not counted as additional zero-emissions energy.

Verifiers must assess that only reported electricity coming from OPS and renewable energy generation onboard are counted as zero-emissions electricity, while electricity generated from fuel is only counted as fuel-based energy (and not again as electricity) with the appropriate emissions factors. To verify the reliability of the reported data, verifiers may take the following example actions:

- Actions related to the delivery of electricity via OPS as established in Section 3.2
- If the ship is equipped with battery storage: the actions outlined above
- Considering whether the output of renewable energy generation onboard is realistic: for example, comparing with typical values (seasonally adjusted, where relevant) or technical documentation of the equipment
- Review maintenance plans and documentation for renewable energy generation equipment, that ensures the equipment was operational when its use is reported
- Request ship information that would allow to clearly establish how the electricity from fossil fuels would be accounted for (e.g. relevant schematics, documentation for metering and switchboards, etc), to establish whether there is a risk of double counting.⁶⁰ Also applicable to review the completeness of the monitoring of electricity generation and consumption
- If there is a risk of double counting electricity generated from fossil fuels: assessing the monitoring of this electricity, and ensuring that it has been excluded from the total electricity consumption of the ship (and only counted as fuel consumption)
- Where applicable: request software logs of energy management systems on ships that can indicate the timestamped generation and use of electricity onboard

3.4 Data reported for applying derogation for sailing in ice conditions

The above has to be reported in Part F of the FuelEU report.

Date, time and position when entering and leaving the ice conditions and the distance travelled when sailing in ice conditions

⁶⁰ Given the higher verification risk previously expressed, it is better if a site visit has been performed during a prior assessment of the monitoring plan, to ensure the monitoring plan and ship documentation match reality.

In order to verify the reliability of the reported data, verifiers might take the following example actions:

- Request logbook entries for the voyage: has time entering/leaving ice conditions been recorded in conformity to the procedure indicated in Table C.3.3 of the FuelEU monitoring plan?
- Rationalise whether sailing through ice at that location and time is realistic: for example, if sailing in ice is reported in the southern Baltic sea in summer this might indicate a reporting error to be scrutinised.
- Consult (historical) data for sea ice for the region and period where the voyages in question took place. Was it likely the ship experienced ice on the voyage at the recorded time of year?
 - Examples of sources of observational ice information are given in OCIMF/ICS [Guidelines for the Development of a Polar Water Operational Manual](#) including the following:
 - [Sea-Ice Information Services in the World](#), (World Meteorological Organization)
 - [Danish Meteorological Institute](#)
 - Ice Watch / Norwegian Meteorological Institute : <https://cryo.met.no/en/icewatch>

Historical data on ice could be compared with (for instance) vessel tracking data to show the route taken by the ship, or records made available by the ship from its ECDIS equipment.

- In cases where the verifier has reasons to question the presence of sea-ice: request a copy of the voyage plan prepared by the master prior to sailing. While conditions at sea are never constant, the voyage plan is, according to the Polar Code⁶¹, to include the following information which might support the verifier to corroborate the reported sailing through ice:
 - current information on the extent and type of ice and icebergs in the vicinity of the intended route
 - statistical information on ice and temperatures from former years;
- The verifier could also decide to request a copy of the ice charts or aids to navigation that were used for the voyage in question. The Polar Code requires that ships have means of receiving and displaying current information on ice conditions in the area of operation⁶². This information could support the verifier to confirm the plausibility of the sea ice that was encountered and reported as sailing through ice. At the same time, it is acknowledged that sea ice is not constant especially in the case of drifting ice so historical ice charts cannot be relied on in full by the verifier⁶³.
- Where applicable, request engine logs or other software which could indicate the ship was operating in ice conditions.
- Where applicable, request information concerning ice-breaking services that were used as part of the journey through ice.

Fuel consumption when sailing in ice conditions:

In order to verify the reliability of the reported data, verifiers might consider the following example actions:

- Is the reported fuel consumption for sailing in ice conditions for a certain voyage by the ship in a similar order of magnitude to reported fuel consumption for a similar voyage in the same time of year or are there variations which cannot be explained?

⁶¹ See Polar Code Part I-A Chapter 11 “Voyage Planning”

<https://wwwcdn.imo.org/localresources/en/MediaCentre/HotTopics/Documents/POLAR%20CODE%20TEXT%20AS%20ADOPTED.pdf>

⁶² See Polar Code Part I-A Chapter “Safety of Navigation”

<https://wwwcdn.imo.org/localresources/en/MediaCentre/HotTopics/Documents/POLAR%20CODE%20TEXT%20AS%20ADOPTED.pdf>

⁶³ The Polar Code itself states that that in many areas, the chart coverage may not currently be adequate for coastal navigation. In these areas especially, determining the location and extent of sea ice may be particularly challenging.

- In cases where the verifier has reasons to question if the ship sailed through ice at the reported time, they may for example request engine room logbook entries or disaggregated data logs showing fuel consumption throughout the duration of the voyage in question. These records/data could then be compared to the distance the ship travelled over the same period: is there evidence of the main engines operating at higher loads to account for the higher fuel consumption, without a corresponding increase in speed?

4 Flexibility mechanisms

The FuelEU Maritime Regulation provides the possibility for the compliance of ships to be managed via so called flexibility mechanisms which are set out in Articles 20 and 21 of the Regulation. The verifier has a role in each of these mechanisms, which begin after the issuance of a ship's FuelEU verification report (by 31 March).

This chapter begins, in section 4.1, with an overview of the rules, conditions, and terminology related to flexibility mechanisms in general. Thereafter, specific rules applicable to each individual mechanism are given in sections 4.2 – 4.4. The main task of the verifier with regard to flexibility mechanisms is to ensure that the rules and conditions of each flexibility mechanism are correctly applied, whether they are applied individually or in combination. Throughout the chapter, key questions and clarifications in this regard are highlighted in text boxes.

Note: Reference is made only to flexibility mechanisms and compliance balance as defined in the FuelEU Maritime Regulation. A ship's compliance balance to an eventual FuelEU RFNBO subtarget from 2031 is also not addressed in this guidance.

4.1 Introduction to flexibility mechanisms

Article 16(5) of the FuelEU Maritime Regulation requires the verifier to notify the shipping company, by 31 March, of the results of their verification of the FuelEU report. This notification includes the issuance of the verification report⁶⁴ and, for ships with a FuelEU report assessed as satisfactory, the calculation of the ship's compliance balance for GHG intensity⁶⁵.

The compliance balance for GHG intensity is, to recall, the difference between the ship's actual GHG intensity and the target GHG intensity for a reporting year, multiplied by the ship's energy in scope, as shown in the formula below. The units are grams of CO₂ equivalent (gCO₂eq).

Compliance balance [gCO ₂ eq] =	$(GHGIE_{\text{target}} - GHGIE_{\text{actual}}) \times [\sum_i^{n_{\text{fuel}}} M_i \times LCV_i + \sum_k^c E_k]$
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Figure 2: Formula for calculating the ship's GHG intensity compliance balance. FuelEU Annex IV Part A

On the basis of the ship's compliance balance, the managing company has several options to manage the ship's compliance:

⁶⁴ Annex II of Implementing Regulation (EU) 2024/2027

⁶⁵ As well as the yearly average GHG intensity of the energy used on board by the ship concerned and the other information in FuelEU Article 16(4).

1. In the case of a negative compliance balance (deficit):

Ships may **borrow** a compliance surplus from the following period or **pool** for compliance with other ships which have a surplus. A third option is to pay a penalty, which may also apply if after the application of flexibility mechanisms there is still a deficit. Penalties are not addressed in this guidance for verification.

2. In the case of a positive compliance balance (surplus):

Ships may **bank** the surplus for use in any subsequent period or make it available in a **pool** for ships which have a compliance balance deficit. A third option is to take no action with the surplus, in which case it will be voided once the FuelEU DoC is issued.⁶⁶ This last option is not addressed in this guidance for verification.

If a company decides to use pooling or borrowing this choice needs to be declared and verified by 30 April. However if the company decides to bank compliance surplus (either the ship's own surplus or surplus gained after pooling) the deadline to declare and have this verified may be later, until the issuance of the FuelEU DoC by 30 June 2026.

4.1.1 Compliance balance terminology:

General terminology applicable to all flexibility mechanisms is defined below, consistent with the terminology used in the compliance balance module of the FuelEU Database⁶⁷. In all cases, compliance balance is in units of grams of CO₂ equivalent.

Initial Compliance Balance (ICB): this is a ship's compliance balance following verification of the FuelEU report by 31 March.

Adjusted Compliance Balance (ACB): this is the ICB +/- compliance surplus/deficit from the previous year. In the case of a banked surplus in the previous year, this surplus is added to the ICB whereas if the ship borrowed compliance balance in the previous year this is subtracted from the ICB. The ACB is determined directly in the FuelEU Database but since 2025 was the first FuelEU reporting period there were no ships with a banked or borrowed compliance amount from the previous period, so ACB was not applicable in the first verification period of 2026

Verified Compliance Balance (VCB): this is the ship's final compliance balance after the application of all applicable flexibility mechanisms (banking, borrowing or pooling)⁶⁸.

⁶⁶ A ship may choose to not bank any surplus, for example, to retire it in the context of participating in voluntary markets. The FuelEU Maritime Regulation and the FuelEU Database are not designed to directly support these. The need to verify or trace the retirement of the surplus and any other needs stemming from participating in these voluntary markets are beyond the scope of the Regulation and need to be considered by private actors participating in these schemes.

⁶⁷ EMSA 6th FuelEU webinar: Compliance Balance Module 4 December 2025 https://www.emsa.europa.eu/reducing-emissions/webinars_and_tutorials/fueleu-webinars/download/8423/5222/23.html

⁶⁸ FuelEU article 23(1)

Figure 3 below presents the different steps in the calculation of the compliance balance with the application of flexibility mechanisms, as explained above. The figure also outlines when the calculation and payment of penalties are due (if the verified compliance balance results in a deficit) to obtain the FuelEU DoC.



Figure 3: Different steps in the calculation of the compliance balance

4.1.2 Prerequisites for flexibility mechanisms in general

The following prerequisites apply to all flexibility mechanisms:

- The ship is in scope of FuelEU
- The ship has, for the concerned reporting year, submitted a FuelEU report to its verifier
- The verifier has subsequently assessed the FuelEU report as “satisfactory” and issued a verification report to the company which is also available on the FuelEU Database
- The ship has not yet been issued a FuelEU document of compliance for the given verification period.

Note that the FuelEU Database will only generate an “Initial Compliance Balance” for ships for which each of the above prerequisites are met.

4.1.3 FuelEU Database compliance balance module as a tool to support verification

The FuelEU Database compliance balance module has been established to support the effective functioning of the Regulation. Certain elements of the module have also been designed to reduce administrative burden, human error and to support the process of verification.

For instance, the FuelEU Database does not require verifiers to manually enter the compliance balance of ships (whether that is the initial, adjusted or verified compliance balance) into the system. Instead, the FuelEU Database updates the compliance balance value based on a FuelEU report that is verified as satisfactory, a ship’s compliance balance from previous years and options selected by shipping companies to, for example, bank compliance surplus.

The FuelEU Database is also set up with a number of system controls to support the correct application of flexibility mechanisms for each ship. For example, the FuelEU Database will not allow a ship to bank more compliance surplus than the ship has available or borrow more compliance balance surplus than is legally permitted. Also, concerning pooling, the system supports the fulfilment of the regulatory rules by only allowing pools to be formed when the sum of the compliance balance of ships in the pool is zero or positive.

The functionality and system controls of the FuelEU Database compliance balance module can therefore be a tool to support verification activities. **However, it is nonetheless emphasised that verifiers are ultimately responsible for ensuring the correct application of the FuelEU rules and related calculations.**

It is worth noting that in case of a ship being transferred to another company, the transferring company needs to prepare a partial FuelEU report, to be verified by the verifier of the transferring company. The receiving company, or the company who has responsibility of the ship by 31 December of the reporting period is responsible for the compliance of the ship, also needing to prepare the FuelEU report, to be verified by their verifier. In this case, the FuelEU Database does not directly allow for visibility of previous partial FuelEU reports, but the company and the verifier should be able to proceed on the basis of the verified partial FuelEU report outcomes.

4.1.4 General questions relating to all flexibility mechanisms

If a ship does not make a port of call within the scope of FuelEU within a reporting period, can the ship still submit a FuelEU report?

Yes. The Regulation does not prohibit a FuelEU report being voluntarily submitted for verification and the eventual issuance of a FuelEU Document of Compliance even if the ship was not active in the scope of the Regulation during a reporting year. This is, in other words, a “zero energy” FuelEU report. This is allowed even for ships who had never had a port call in a port covered by the FuelEU Maritime Regulation, to facilitate compliance (i.e. to obtain a FuelEU DoC) in case of an eventual port call under the scope.

The implications of this on flexibility mechanisms include the following:

- **With regards to pooling and banking:** In the case of a ship with a banked compliance surplus from previous years: the compliance surplus may be used in a pool in the current verification period provided the ship has a FuelEU DoC valid for the reporting year subject to verification. However, a ship which has never had a port call in a port covered by the Regulation may not join a pool, even if it has submitted a zero energy FuelEU report and holds a FuelEU DoC, as formally this ship has not been covered by the Regulation,
- **With regards to borrowing:** Article 20(4) states that where a ship does not have any port call in the Union during the reporting period and borrowed an advance compliance surplus in the previous reporting period, the administering State can issue a penalty corresponding to the amount that was initially avoided, multiplied by 1,1. The above interpretation however means that this penalty would not be applicable if the ship obtains a FuelEU DoC for the present verification period.

If a ship is scrapped or is a total loss during a year, can that ship still apply flexibility mechanisms?

If a ship which was active by 1 January of the reporting period was rendered permanently out of commission for any reason before the end of the year, the ship is still subject to all the responsibilities and obligations under the FuelEU Maritime Regulation (if under the scope), and may participate in flexibility mechanisms to comply with them.

This means that the company who was last responsible for the operation of the ship must still submit a FuelEU report for verification if the ship made any voyages under the scope while it was still active, and is liable to ensure that the energy used onboard complied with the requirements set by the Regulation. If the ship has a deficit in the adjusted compliance balance, it may still choose to

participate in a pool and/or pay the penalty for the outstanding deficit (but logically may not choose to borrow, as there will be no following period).

If such a ship did not make any voyages under the scope of the Regulation, the company may still choose to voluntarily submit a zero energy FuelEU report. This can be done, for example, with the purpose of resolving any deficit from previously borrowed compliance through flexibility mechanisms or offering any remaining banked surplus in a pool.

Should the verifier consider a ship's Adjusted Compliance Balance (adjusted on the basis of previous banking/borrowing) as final when it is recorded in the FuelEU Database?

Yes. The ACB is determined in the FuelEU Database on the basis of any verified compliance balance surplus/deficit for the ship from the previous verification period, without a need for manual input from the verifier.

The verifier should consider a ship's Adjusted Compliance Balance as final when recorded in the FuelEU Database, also in the case of a change of verifier between reporting periods. There is no obligation for the new verifier to review the activities or calculations carried out by the previous verifier.

Note also that, in the context of competent authorities' right to conduct additional checks on FuelEU reports, verification, or calculations (Article 17), a compliance balance which has been verified – and from which a FuelEU DoC has been issued - will not be subsequently changed in the event of discrepancies found. This is to avoid propagating any discrepancies to further years, or to pools where a ship may have participated. Instead, competent authorities may issue an additional penalty to account for the difference, if needed.

4.2 Borrowing

Where a ship has a compliance deficit (i.e. a negative adjusted compliance balance) and the company decides to borrow an advance compliance surplus, verifiers are – according to article 20(3) required to approve that borrowing is correctly applied. According to article 28(3) of Implementing Regulation (EU) 2024/2027, the verifier should in particular ensure that the advance compliance surplus multiplied by 1.1 is subtracted from the same ship's compliance balance in the following reporting period⁶⁹ and that the borrowing adheres to the rules of article 20(2) of the FuelEU Regulation.

The rules for borrowing are as follows:

- Article 20(2)(a): The advance compliance surplus that is borrowed cannot exceed 2% of the GHG intensity target for the period in which the ship reported, multiplied by the ship's total energy consumption. Note: the so called "borrowing-limit" is generated automatically in the FuelEU Database as a supporting tool. In cases where a ship's borrowing limit exceeds the magnitude of the deficit in a ship's ICB (or, after the first verification period, the ACB) the FuelEU Database does not allow for borrowing to be selected.
- Article 20(2):
 - A ship cannot borrow only part of the compliance deficit⁷⁰; it must borrow the full amount (when borrowing is possible) or not borrow at all. This for example means that a ship

⁶⁹ This is the calculation of the adjusted compliance balance in the following period

⁷⁰ FuelEU article 20(2) states that in the case of a compliance deficit, "the company may borrow an advance compliance surplus of the corresponding amount from the following reporting period."

- cannot borrow part of its deficit and pay a penalty or join a pool to cover the rest of the deficit.
- The advance compliance surplus that is borrowed can only be borrowed from the reporting period which directly follows⁷¹; it is not possible to borrow more than 1 reporting period in the future.
- Article 20(2)(b): A ship may not borrow an advance compliance surplus for two consecutive reporting periods.
- Article 21(7): A ship cannot borrow advance compliance surplus and then join a pool in the same verification period. To control this, the FuelEU Database disables the option for ships to join a pool once borrowing has taken place.

Note that for the verification period following one in which borrowing took place, the **Adjusted Compliance Balance** will be automatically recorded in the FuelEU Database. It will subtract not only the borrowed amount but also an additional 10% “interest rate” in accordance with FuelEU article 20(2). The FuelEU Database assists by carrying out this calculation without additional input from the verifier.

4.2.1 Verification questions relating to borrowing

How does a verifier ensure that a ship does not borrow an advance compliance surplus for two consecutive periods?

The FuelEU Database compliance balance module shows the previous advance compliance surplus including aggravation, in addition to an automatic calculation of the adjusted compliance balance. In practice, if the value of the previous advance compliance surplus including aggravation is different than zero, this means that the ship borrowed compliance in the previous period, and thus is not eligible to borrow for a second consecutive period. The FuelEU Database will also disable the option for borrowing if it has been done in the previous period.

4.3 Pooling

According to Article 21 of the FuelEU Regulation, verifiers are required to approve pooling arrangements for the purpose of compliance with Article 4, which sets the Regulation’s GHG intensity limits, and, where applicable, Article 5(3), which may require 2% of a ship’s annual energy to come from RFNBOs from 2034. This document addresses pooling only in relation to compliance with the GHG intensity limits under Article 4. Separate guidance would be needed if Article 5(3) and the associated RFNBO subtarget become applicable.

Pooling operates by summing and redistributing⁷² the compliance balances of ships within a pool, subject to agreement by the managers of the ships concerned and verification by an appointed pool verifier. The compliance balances of the ships are summed so that ships with a surplus can compensate those with a deficit. For a pool to be valid under FuelEU, the total pool compliance balance must be positive.

The pool verifier must confirm that the pool composition complies with the rules set out in the Regulation and that this is correctly recorded in the FuelEU database.

⁷¹ FuelEU article 20(2) states “the advance compliance surplus may not be borrowed”. It follows that borrowing is only permitted from the reporting period directly after the one currently subject to verification.

⁷² Article 3(38) states: ‘total pool compliance balance’ means the sum of the compliance balances of all ships included in the pool

The Regulation establishes the following requirements for pooling in Article 21:

- A ship's compliance balance may not be included in more than one pool during the same reporting period (Article 21(1))
- Ships that have borrowed an advance compliance balance surplus from the following reporting period may not participate in a pool (Article 21(7))
- A pool is valid only if, under Article 21(4):
 - the total pool compliance balance is positive;
 - ships that originally had a compliance deficit do not end up with a higher compliance deficit after pooling;
 - ships that originally had a compliance surplus do not end up with a deficit after pooling.

The Regulation does not impose any further specific limits on pooling. In particular:

- There is no limit to the number of ships in a pool
- There is full flexibility in the ships in scope of the Regulation which may be pooled together e.g. regardless of ship type, owner, ISM company, flag state, etc

Under Article 21(6), if an individual ship retains a compliance surplus after pooling, that surplus may be banked for the following reporting period in accordance with Article 20(1). However, banking is only possible before the FuelEU Document of Compliance is issued, which must be done no later than 30 June. Where surplus compliance is banked after pooling, this requires approval from the ship verifier.

In practice, the pool verifier must confirm the following:

- each ship's compliance balance is included in only one pool;
- the total pool compliance balance, calculated as the sum of the adjusted compliance balances of all ships in the pool, is positive;
- the allocation of the total compliance balance is arithmetically correct;
- the sum of the compliance balances before pooling is positive or zero.

FuelEU Database includes system controls to check some of these requirements, including that the total compliance balance of the participating ships is positive or zero before pooling redistributions and that the total compliance balance is equal before and after pooling. However, the pool verifier remains responsible for confirming that all applicable requirements are satisfied.

4.3.1 Verification questions relating to pooling

How does a pool verifier ensure that a ship is only included in one pool?

The intention of a ship to form part of a pool must be registered in the FuelEU Database. As such, the FuelEU Database supports pool verifiers by providing them visibility as to whether the ship is already included in a pool for the relevant compliance balance. The FuelEU Database would not allow a ship to be included in a second pool for the same compliance balance.

Note: Two separate pools may be used for GHG intensity target and (if applicable in the future years) for the subtarget for RFNBO.

How does a pool verifier check that a ship has not borrowed an advance compliance surplus from the following reporting period?

A ship cannot borrow an advance compliance surplus and participate in a pool in the same period (Article 21(7)). The pool verifier must ensure that a ship in a pool is not borrowing an advance compliance surplus. The FuelEU Database also assists in this regard, by blocking the pooling option if a ship is borrowing in the same period, so the system will not allow it.

How should a pool verifier understand the obligation to hold a valid FuelEU Document of Compliance when the ship has not been previously a subject of the FuelEU Maritime Regulation?

The obligation to hold a valid FuelEU document of compliance as defined in Article 24 is necessary for a ship participating in a pool. This obligation applies by 30 June of the verification period.

However, ships that have not been in the scope of the FuelEU Maritime Regulation in previous years, do not necessarily hold a valid FuelEU DoC. Ships that fall under the scope of the FuelEU Maritime Regulation for the first time in the current period are not required to hold a valid DoC. These ships may join a pool the first time they fall under the scope. The verifier should make the required assessment to verify to a reasonable assurance the ship is compliant with the regulation regarding previous years, if it is being claimed by the company as being out of scope.

A ship which has never been under the scope of the FuelEU Maritime Regulation (i.e. not made any voyages to/from a port of call covered by the Regulation) at least once since 2025 may not join a pool, regardless of whether they obtained a FuelEU DoC by means of submitting a zero energy FuelEU report on a previous period. Pool verifiers can see whether a ship has been under the scope of the Regulation by means of checking the compliance balance module in the FuelEU Database.

Ships that previously fell under the scope of the FuelEU Maritime Regulation, but are not in the current period and hold a FuelEU DoC, may still join a pool by submitting a zero energy FuelEU report.

What happens if, after a pool has been registered and verified, it is found that the compliance balance of a ship before entering the pool was incorrectly calculated?

Once the pool outcome is verified, it can still be revised if no ship in the pool has a verified compliance balance (i.e. the compliance balance is in draft or under revision status for all ships in the pool).⁷³ This allows for a different distribution of the pooled compliance balance to be proposed, including if a mistake were to be found in the compliance balance of a ship before entering the pool.

It should be noted that the pool verifier has no obligation to assess the calculations done by the individual ship verifiers, nor address any mistakes found in individual ship verification reports, only to ensure that the ships have been verified by an accredited verifier and that pool rules are complied with. This means that the pool verifier can use the results of the ship verification (i.e. the adjusted compliance balance) with that purpose, without otherwise needing to check the verification report. Mistakes made in the individual ship verification report are the responsibility of the ship verifier.

If, however, a mistake is found after the pool is verified and at least one ship in the pool has already logged a verified compliance balance in the FuelEU Database, the pool can no longer be revised.⁷⁴ If the ship which had the mistake has a compliance balance deficit that was not fully compensated in the pool, it may be subject to a penalty. This is also the case if competent authorities find discrepancies in the verified compliance balance during their additional checks, for which they may choose to correct the discrepancies through the issuance of a corresponding penalty, if applicable.

⁷³ It is possible, however, that due to short timelines this is not possible in practice.

⁷⁴ This is to prevent a discrepancy or mistake from having consequences that propagate over different ships in a pool and over multiple reporting periods.

If instead the correction of the mistake would lead to a (higher) surplus, the ship will not be subject to a penalty, but no specific process exists to then allow the ship to bank that surplus.

4.4 Banking

For ships with a compliance surplus, companies may decide to bank this compliance surplus for future reporting periods.

Compliance surplus can be earned:

- From the ship's actual compliance balance as stated on its verification report
- As a result of pooling, in the case of a ship leaving a pool with a compliance surplus

The banking of compliance surplus can only take place before the FuelEU document of compliance is issued by the verifier.

According to article 20(1) of the FuelEU Regulation, verifiers are responsible for approving a shipping company's application of banking of compliance surplus. To do so, according to Implementing Regulation (EU) 2024/2027, the verifier should – following a request from a shipping company to bank the compliance surplus, verify that the surplus is in accordance with FuelEU articles 19, 20 and 21 and the calculations referred to in article 26 of Implementing Regulation (EU) 2024/2027. Where a compliance surplus results from the total pool compliance balance, Implementing Regulation (EU) 2024/2027 article 28(2) requires that the verifier also assess the calculations and the information related to the pooling of compliance, before approving the banking of the compliance surplus.

4.4.1 Questions relating to banking

In the event of banking after pooling, to what extent is the ship verifier responsible for checking the verification activities and calculations of a different pool verifier?

At what stage of the verification period should the compliance surplus of a ship be banked?

Banking of compliance surplus can be requested by the company before the issuance of the FuelEU DoC. Within the same verification period, a ship can both bank a compliance surplus and partake in a pool. In this case, banking can be done after the pool is verified and surplus allocated but must be done before issuance of the FuelEU DoC. In this case, ships participating in a pool need to a verified complied balance (after pooling) by 30 April of the verification period. The verifier can then ask the company whether they intend to bank any surplus, and if it is the case, register the banking before issuing the FuelEU DoC by 30 June.

It is recommended that the "pool assessment" document, available from the FuelEU Database after a satisfactory verification by the pool verifier, is shared with the ship verifier. While the ship verifier has no obligation to assess the calculations and the information related to the pool, but has to confirm that the pool was verified, before approving the banking of compliance surplus.

The decision to bank rests with the company. Upon receiving the company's request, the verifier will process it within the FuelEU Database Compliance Balance Module.

According to the FuelEU Guidance Document, the banked compliance surplus does not have an expiration date, how does this look in practice?

Contrary to the borrowing of compliance, banking can be done for an indefinite number of consecutive periods. This means that a ship which has obtained a compliance surplus in consecutive years (and not spent it, for example, in a pool) can keep banking said surplus.

In practice, this means that the surplus needs to be banked by companies in the FuelEU Database. For example, if a ship has a compliance surplus for a number of years the surplus of the first year may be banked. In the following year, the surplus of the first year will be accounted for in the adjusted compliance balance in addition to any surplus of that year, the total of which may be banked again.

There may be cases where a ship has a banked compliance surplus from a previous period (when the ship was active under the scope of the Regulation) but does not make any voyages/port activity falling under the scope of the Regulation in a subsequent reporting period. In such cases, in order to keep the surplus banked, the ship is not obliged to submit a zero-energy FuelEU report each year - unless the shipping company intends to make that surplus available for pooling in a given year.

It is therefore possible that, for a ship with a banked compliance surplus, one or more reporting periods occur without the submission of a FuelEU report, including a zero-energy FuelEU report. In such situations, when the ship subsequently calls at a port under the scope of the Regulation (thereby triggering a reporting obligation) or submits a zero-energy FuelEU report (e.g. for pooling purposes), the verifier needs to confirm the status of the preceding period(s) without reports. This includes verifying that no port calls under the scope of the Regulation took place during those periods and that no reporting obligations were missed. This ensures that the banked compliance surplus is accurate.